

Effect of Central Nodes on Task Allocation in Distributed Satellite Systems for Wildfire Detection

Communication-Constrained Dissemination, Follow-Up Observation, and Architecture Trade-Space Analysis

Maneesh Manjunath

Thesis for the attainment of the academic degree

Master of Science (M.Sc.)

at the TUM School of Engineering and Design of the Technical University of Munich.

Examiner:

To be confirmed

Advisor:

Vincenzo Messina

Submitted:

Munich, Working draft, 2 September 2026

I hereby declare that this thesis is entirely the result of my own work except where otherwise indicated. I have only used the resources given in the list of references.

Draft Status and Reading Guide

Working draft for supervisor review, updated on 2 September 2026. This version integrates every locally available validated evidence layer while keeping unfinished work visible. It is not a submission copy.

The report uses two related campaigns that must not be numerically merged:

1. the validated two-region nominal campaign, containing 891 Walker/CN cases for California and 891 for India, is retained for the regional comparison and the historical nominal trade-space result; and
2. the newer *Full891 Fresh V2* campaign, containing 539 scenario definitions per architecture case, is the authoritative source for the updated simulator, California policy ablation, controlled task-size experiments, injected failures, and revised objective-weight analysis.

The California Fresh V2 archive contains all 891 architecture identifiers. A total of 890 cases have complete success markers. The remaining case, T180_P10_H1000_I0986_CN100_F1, contains 400 of 539 expected scenario rows and is excluded from comparisons that require a complete scenario set. After removing 4,269 resume duplicates, 480,110 unique scenario rows remain. All 19,179 recorded architecture-validation checks pass and no completed scientific row has status FAILED.

No completed India Fresh V2 archive was found in the accessible project data at the time of this update. India-specific robustness and controlled-sensitivity values are therefore not inferred from California or from the earlier nominal campaign. The existing two-region nominal results remain identified as a separate evidence layer. Observation-radius sensitivity has not been executed, and Fresh V2 does not retain a separately named first-reception-latency metric in its canonical case table; these limitations are stated wherever they affect a research question.

The official submission date, examiner, declaration, and final repository release tag remain administrative open items. The source and compiled PDF in this repository are intended for transparent supervisor review and automated checking.

Zusammenfassung

Verteilte Satellitensysteme können zeitkritische Erdbeobachtungsaufgaben schneller bearbeiten, ihre Leistungsfähigkeit hängt jedoch von orbitalem Zugang, begrenzten Kommunikationsfenstern und der gewählten Weiterleitungsstrategie ab. Diese Arbeit untersucht den Einfluss eines Zentralen Knotenanteils auf die Verteilung von Waldbrandaufgaben, die nachfolgende Beobachtung, den Ressourcenbedarf und die Empfindlichkeit gegen Ausfälle. Eine nachvollziehbare Datenpipeline erzeugt aus deduplizierten Sentinel-3-SLSTR-FRP-Detektionen 805 priorisierte Aufgaben: 32 für Kalifornien und 773 für Indien.

Der Entwurfsraum umfasst 60, 120 und 180 Satelliten, 3, 5 und 10 Bahnebenen, Höhen von 500, 800 und 1000 km, Inklinationen von 60, 80 und 98,6 Grad sowie Zentralen Knotenanteile zwischen 0 und 100 Prozent. Die frühere nominale Kampagne enthielt 891 validierte Fälle pro Region. Die neuere Full891-Fresh-V2-Kampagne definiert 539 Betriebs-, Policy-, Aufgabengrößen- und Fehler-szenarien pro Architektur. Für Kalifornien liegen 890 vollständige Fälle und nach Entfernung von Resume-Duplikaten 480.110 eindeutige Szenariozeilen vor; alle 19.179 Architekturprüfungen wurden bestanden.

In der begrenzten nominalen Policy beträgt die mittlere Beobachtungserfüllung 61,59 Prozent, die strikte vollständige Verteilung an nutzbare Empfänger 0,63 Prozent und deren mittlere Abdeckung 15,29 Prozent. Das diagnostische Szenario ohne Hop- und Kopienbegrenzung erhöht diese Werte auf 77,62, 74,69 und 91,14 Prozent, steigert jedoch die mittlere übertragene Datenmenge von 197,55 auf 2.047,57 Mbit pro Fall. Die Wirkung des Zentralen Knotenanteils ist nicht monoton und zeigt abnehmenden Grenznutzen bei weiter wachsender Kommunikationslast.

Bei einem Stressniveau von 30 Prozent ist der Ausfall der einzigen Bodenstation die dominante kalifornische Schwachstelle. Eine Vervierfachung der Aufgabengröße reduziert die mittlere Beobachtungserfüllung von 77,62 auf 69,48 Prozent. Eine Pareto-konsistente Gewichtungsanalyse bevorzugt bei ausgewogenen Annahmen eine Konfiguration mit 60 Satelliten, 10 Ebenen, 800 km Höhe, 98,6 Grad Inklination und 40 Prozent Zentralen Knoten. Diese Konfiguration gewinnt 41,79 Prozent von 10.000 zufälligen Gewichtungsziehungen. Das Ergebnis ist ein bedingter Kompromiss und kein universelles Optimum. Die Indien-Fresh-V2-Ergebnisse und eine Sensitivität des Beobachtungsradius bleiben vor der abschließenden regionsübergreifenden Robustheitsaussage offen.

Abstract

Distributed satellite systems can shorten the response to time-critical Earth-observation events, but their performance depends on orbital access, finite communication opportunities, and the routing policy used to disseminate tasks. This thesis studies how assigning a fraction of a Walker constellation to a central-node communication and routing role affects wildfire-task dissemination, follow-up observation, resource demand, and failure sensitivity. A traceable preprocessing pipeline converts de-duplicated Sentinel-3 SLSTR fire-radiative-power detections into 805 prioritized tasks: 32 for California and 773 for India. Each task retains a release time, location, priority-dependent data volume, deadline, and source provenance.

The architecture grid combines 60, 120, and 180 satellites; 3, 5, and 10 planes; altitudes of 500, 800, and 1000 km; inclinations of 60, 80, and 98.6 degrees; and central-node fractions from 0 to 100 percent. The earlier nominal campaign contains 891 validated cases per region and establishes that architecture preferences are region- and metric-dependent. The newer Full891 Fresh V2 campaign evaluates 539 operational, policy, task-size, and failure scenarios per architecture. The locally available California archive contains 890 complete cases and 480,110 unique case-scenario rows after resume de-duplication; all 19,179 architecture checks pass.

For California Fresh V2, the bounded nominal policy achieves mean observation fulfilment of 61.59 percent, strict useful-recipient dissemination of 0.63 percent, and useful-recipient coverage of 15.29 percent. Removing hop and copy restrictions in the `unlimited_useful_deadline` diagnostic increases these means to 77.62, 74.69, and 91.14 percent, at the cost of increasing transferred data from 197.55 to 2,047.57 Mbit per case. Under the nominal policy, observation fulfilment is non-monotonic: it reaches 71.13 percent at 50 percent central nodes in the balanced subset and falls to 35.70 percent at 100 percent. Under the unlimited diagnostic, dissemination saturates while communication burden continues to rise.

At a 30 percent stress level, ground-station outage is the dominant California vulnerability, reducing observation and strict dissemination by 21.46 and 20.84 percentage points on average. Random satellite and central-node failures produce smaller mean losses, while capacity degradation has limited average effect. Increasing task size from one to four times the nominal value reduces mean observation fulfilment from 77.62 to 69.48 percent and strict dissemination from 74.69 to 60.10 percent. A Pareto-consistent objective-weight analysis selects a 60-satellite, 10-plane, 800 km, 98.6-degree, 40-percent-central-node configuration for balanced and performance-dominant preferences; it wins 41.79 percent of 10,000 randomized preference draws. This is a conditional compromise, not a universal optimum. India Fresh V2 and observation-radius sensitivity remain required before the final cross-region robustness conclusion.

Acknowledgements

I thank Vincenzo Messina for providing the communication model, facilitating the use of TAT-C, and guiding the development of the research questions and evaluation requirements. I am grateful to the Chair of Spacecraft Systems at the Technical University of Munich for the academic environment in which this work was conducted.

The wildfire-data pipeline, integration code, campaign automation, post-processing, analysis, and report preparation were carried out by the author. The communication model and the TAT-C basis were provided by Vincenzo Messina and are identified as such in the methodology and contribution statement. This wording should be reviewed before the final submission together with the official authorship and software-licensing statements.

Contents

Draft Status and Reading Guide	iii
Zusammenfassung	iv
Abstract	v
Acknowledgements	vi
List of Figures	xi
List of Tables	xiii
Acronyms	xiv
1 Introduction	1
1.1 Motivation	1
1.2 Problem Statement	1
1.3 Aim and Objectives	2
1.4 Research Questions	3
1.5 Hypotheses	3
1.6 Scope and Boundaries	3
1.7 Meaning of a Central Node	4
1.8 Scientific and Engineering Contributions	4
1.9 Report Structure	5
2 Background and Related Work	6
2.1 Distributed Satellite Systems	6
2.2 Task Allocation and Observation Scheduling	6
2.3 Time-Varying Satellite Connectivity	7
2.4 Central Nodes as a Hybrid Architectural Mechanism	8
2.5 Wildfire Detection from Space	8
2.6 From Fire Pixels to Communication Tasks	9
2.7 TAT-C and Supporting Orbital Tools	10
2.8 Synthesis of Related Work and Thesis Positioning	10
2.9 Research Gap	11
3 Wildfire Task Dataset and Preprocessing	12
3.1 Role of the Wildfire Dataset	12
3.2 Event Windows and Regional Roles	12
3.3 Processing Pipeline	12
3.3.1 Product and NetCDF selection	12
3.3.2 FRP threshold and clustering	14
3.4 Confidence Harmonization	14
3.5 Priority Score	15
3.6 Final Task Population	15
3.7 DSS Task Schema	17

3.8	Validation and Limitations of the Task Dataset	17
4	System Model and Methodology	18
4.1	Methodological Overview	18
4.2	Task Input and Temporal Horizon	18
4.3	Walker Constellation and Central-Node Assignment	18
4.4	Orbit and Observation Opportunity Model	19
4.5	Communication Opportunities and Link Capacity	19
4.6	Topology and Temporal Routing	20
4.7	Scenario Transformations	20
4.8	Outcome and Resource Metrics	20
4.9	Checkpointing, Parquet Output, and Deterministic Resume	21
4.10	Validation before Inference	21
4.11	Pareto and Objective-Weight Method	22
4.12	Implementation Ownership and Method Boundaries	22
5	Experimental Design and Reproducibility	23
5.1	Two Evidence Layers	23
5.2	Common Architecture Grid	23
5.3	Earlier Two-Region Nominal Campaign	23
5.4	Fresh Full891 V2 Scenario Matrix	24
5.5	California Fresh V2 Integrity Audit	24
5.6	Validation Gates and Claim Freeze	26
5.7	Analysis Populations	26
5.8	Output Contract and Post-Processing Inputs	26
5.9	Statistical and Decision Analysis	27
5.10	Compute Configuration and Resume Semantics	27
5.11	Evidence Still Missing	27
6	Validated Two-Region Nominal Baseline	28
6.1	Evidence-Layer Boundary	28
6.2	Result Status	28
6.3	Regional Campaign Summary	28
6.4	Central-Node Fraction Effects	29
6.4.1	Observation fulfilment	29
6.4.2	Useful-recipient completion and receiver coverage	30
6.4.3	First-reception latency	30
6.5	Architecture Factor Effects	31
6.6	Exploratory Correlation	31
6.7	Exploratory Regression of Central-Node Marginal Means	32
6.8	Paired Regional Comparison	33
6.9	Exact Pareto Sets	33
6.10	Nominal Result Synthesis	35
7	Full891 Fresh V2 California Results and Sensitivity	36
7.1	Scope of the Fresh V2 Results	36
7.2	Bounded Policy versus Architecture Potential	36
7.3	Central-Node Fraction	36
7.4	Architecture Factors and Dependence	37
7.5	Policy Ablation	38
7.6	Task-Size Sensitivity	39
7.7	Priority-Class Outcomes	39

7.8	Failure Robustness	41
7.9	Pareto Trade Space	42
7.10	Objective-Weight Analysis	43
7.11	California Result Synthesis	44
8	Discussion	45
8.1	Interpretation Strategy	45
8.2	RQ1: Adapting Distributed Allocation to Real Wildfire Tasks	45
8.3	RQ2: Workload, Priority, and Region	45
8.4	RQ3: Effect of the Central-Node Fraction	46
8.5	RQ4: Best Performance–Complexity–Robustness Trade-Off	46
8.6	RQ5: Stability under Assumptions and Weights	46
8.7	Why the Objective Weights Are Defensible but Not “Discovered”	47
8.8	Policy, Geometry, and Objective Co-Design	47
8.9	Threats to Validity	47
8.9.1	Construct validity	47
8.9.2	Internal validity	48
8.9.3	External validity	48
8.9.4	Statistical conclusion validity	48
8.10	Supervisor-Requirement Traceability	48
8.11	Implications for Thesis Completion	48
9	Conclusions and Outlook	50
9.1	Conclusions	50
9.2	Answers to the Research Questions	51
9.3	Scientific Contribution	51
9.4	Required Work before Final Submission	51
9.5	Outlook	52
A	Detailed Metric Definitions	53
A.1	Notation and Denominators	53
A.2	Task-Level Outcomes	53
A.3	Communication and Network Metrics	54
A.4	Robustness Effects	54
A.5	Pareto Dominance and Normalization	54
A.6	Objective Utility	55
B	Campaign Configuration and Case Identifiers	56
B.1	Case Identifier Grammar	56
B.2	Fresh V2 Frozen Controls	56
B.3	Scenario Counts	57
B.4	Seed Derivation	57
B.5	Checkpoint and Completion Rules	57
B.6	Representative Parquet Reading Commands	57
C	Validation and Evidence Traceability	59
C.1	Evidence Hierarchy	59
C.2	Earlier Nominal Validation	59
C.3	Fresh V2 California Validation	59
C.4	Configuration Fingerprints	59
C.5	Claim Ledger	60
C.6	Automated Report Checks	60

C.7	Reproducibility Boundary	60
D	Supplementary Result Tables	62
D.1	Complete Central-Node Marginal Table	62
D.2	Fresh V2 California Scenario Summary	62
D.3	Fresh V2 California Threshold Counts	62
D.4	Fresh V2 Priority Summary	62
D.5	Objective-Weight Transitions	62
D.6	Factor-Level Means	62
D.7	Metric-Specific Extreme Cases	62
D.8	Pareto Summary	62
D.9	Regression Comparison Table	65
E	Software and Data Availability	67
E.1	Report Repository	67
E.2	Repository Layout	67
E.3	Build and Check Commands	67
E.4	Evidence Files Included	67
E.5	Raw Result Availability	68
E.6	Security and Data Hygiene	68
E.7	Attribution and Reuse	68
	Bibliography	69

List of Figures

3.1	Final de-duplicated Sentinel-3 wildfire-task centroids for California and India, coloured by priority class. The figure is generated directly from the 805-row task input.	13
3.2	Sentinel-3 wildfire-task generation and export workflow.	13
3.3	Absolute regional workload and priority composition of the final task set.	16
4.1	Fresh Full891 V2 simulation and post-processing workflow.	18
5.1	California Fresh V2 completeness and scenario-family coverage after deterministic de-duplication.	25
6.1	Regional marginal response to central-node fraction across the full final grid. S_{100}^{useful} denotes complete dissemination to all simulator-defined useful recipients.	29
6.2	Exploratory regression of the 11 regional central-node marginal means. The displayed model minimizes leave-one-level-out RMSE among linear, quadratic, cubic, and exponential-with-offset candidates.	32
6.3	Paired observation fulfilment for identical architecture identifiers under the California and India task sets.	34
6.4	Observation fulfilment versus non-monetary resource burden. Black outlines identify cases on the exact five-objective Pareto set.	34
7.1	Central-node fraction response across 80 balanced California Fresh V2 base architectures. Shaded or interval displays summarize variation across base geometries. . . .	37
7.2	California Fresh V2 marginal architecture effects under the unlimited useful-deadline replay.	38
7.3	Policy ablation: performance gain and communication burden under central, peer, and hybrid link use.	39
7.4	Task-size sensitivity of performance and communication burden.	40
7.5	California task-priority outcomes under nominal and unlimited scenarios.	40
7.6	Paired change in strict useful dissemination across Fresh V2 robustness families and severities.	41
7.7	California unlimited performance–resource trade space and exact Pareto membership. . . .	42
7.8	Winner transitions as the high-level performance weight λ changes from zero to one. . .	43
7.9	Winner frequency across 10,000 randomized objective-weight draws (seed 20260825). . . .	44

List of Tables

1.1	Objectives, evidence products, and status in this working draft.	2
1.2	Hierarchy of the implemented and planned thesis scope.	4
2.1	Division of roles between FIRMS VIIRS and Sentinel-3 SLSTR.	9
2.2	Related-work synthesis and positioning of the thesis.	11
3.1	Sentinel-3 pipeline refinement and regional total FRP reported during development. .	14
3.2	Final de-duplicated Sentinel-3 task population used by the nominal campaign.	15
3.3	Priority-dependent response deadlines and task packet sizes in the final nominal cam- paign.	16
3.4	Core task fields transferred from the Sentinel-3 pipeline to the DSS simulator.	17
5.1	Result layers and permitted use.	23
5.2	Architecture factors in the Full891 design.	23
5.3	Fresh V2 scenario contract for each architecture case.	24
5.4	California Fresh V2 campaign integrity.	25
6.1	Region-normalized summary across all 891 cases per event region.	28
6.2	Selected regional central-node marginal means across 81 base architectures per row. .	30
6.3	Selected factor-level marginal effects across the full nominal grid.	31
6.4	Selected exploratory Spearman associations across 891 cases per region.	32
6.5	Best exploratory regression model by leave-one-central-node-level-out error.	33
6.6	Spearman rank agreement between California and India over 891 paired architecture identifiers.	33
7.1	California mean outcome over 890 complete Fresh V2 architectures.	36
7.2	Selected unlimited useful-deadline central-node marginal means.	37
7.3	Mean California policy-ablation outcome over 890 complete architectures.	38
7.4	California task-size sensitivity over 890 complete architectures.	39
7.5	Mean paired performance change at the 30-percent stress level.	41
7.6	Illustrative California Fresh V2 Pareto candidates.	42
7.7	Declared weight profiles and selected California architecture.	43
8.1	Status of the agreed analysis requirements.	48
A.1	Metric notation used in the Fresh V2 analysis.	53
B.1	Example case identifiers.	56
B.2	Core Fresh V2 physical and operational controls.	56
B.3	Scenario rows per architecture.	57
C.1	Evidence hierarchy used to resolve conflicting artefacts.	59
C.2	Fresh V2 California validation ledger.	61
C.3	Supported and excluded Fresh V2 claims at the report freeze.	61
D.1	Complete regional central-node marginal results.	63
D.2	Fresh V2 California anchor and policy scenarios (890 complete cases).	63

D.3	Number of complete California architectures meeting selected thresholds.	63
D.4	Mean task-class outcomes over 890 California architectures.	64
D.5	California factor-level marginal means.	64
D.6	India factor-level marginal means.	64
D.7	Metric-specific extreme cases in the final nominal campaign.	65
D.8	Exact Pareto-set size under the declared five-objective vector.	65
D.9	Exploratory regression model comparison.	66

Acronyms

API application programming interface.

CDF cumulative distribution function.

CN central node.

CSV comma-separated values.

DBSCAN Density-Based Spatial Clustering of Applications with Noise.

DSS distributed satellite system.

EO Earth observation.

FIRMS Fire Information for Resource Management System.

FRP fire radiative power.

GS ground station.

HPC high-performance computing.

ISL inter-satellite link.

MW megawatt.

PBL planetary boundary layer.

SLSTR Sea and Land Surface Temperature Radiometer.

TAT task-allocation time.

TAT-C Trade-space Analysis Tool for Constellations.

UTC Coordinated Universal Time.

1 Introduction

1.1 Motivation

Earth-observation missions are moving from isolated spacecraft operations toward constellations in which sensing, communication, and decision-making are distributed across many nodes. This transition creates two coupled design problems. The first is geometric: a spacecraft must obtain access to a target at an operationally useful time. The second is informational: the task describing that target must reach a suitable spacecraft early enough for the access to be used. A constellation with excellent coverage can therefore fail a time-critical mission if the task is disseminated too late, while a well-connected network cannot observe an event for which no suitable access exists.

Wildfire response makes this coupling visible. A remote-sensing product may indicate a new or growing fire, but the detection alone is not the end of the operational chain. The event must be transformed into a task, prioritized, released into the network, forwarded within finite communication opportunities, received by one or more candidate observers, and finally executed before a response deadline. A useful performance evaluation must preserve this event order. It must also distinguish the time to first reception, the breadth of dissemination, and the completion of a follow-up observation.

One way to improve coordination is to assign selected spacecraft a central-node role. Conceptually, a central node may have additional processing, storage, cross-link capability, routing responsibility, coordination authority, or connectivity to ground. Such a node can accelerate information exchange, but it can also increase hardware differentiation and create operational dependencies. The design question is therefore not simply whether central nodes help. It is whether a particular degree of centralization improves mission fulfilment sufficiently to justify the added resource burden and the possible loss of robustness.

This thesis investigates that question through a full-factorial Walker-constellation campaign driven by real, de-duplicated Sentinel-3 wildfire tasks. The implemented experiment varies the fraction of satellites labelled as central nodes while also varying constellation size, plane count, altitude, and inclination. The result is a controlled architecture database in which communication, reception, observation, and resource metrics can be evaluated together.

1.2 Problem Statement

The starting task-allocation logic was developed for a different Earth-observation application. Adapting it to wildfire detection required more than changing the target coordinates. The wildfire use case introduces multiple simultaneous events, unequal priority, priority-dependent deadlines, priority-dependent task sizes, region-specific task density, and the need to validate tasks against physical fire observations. It also requires the simulator to distinguish dissemination from observation fulfilment.

The central problem addressed by this work is stated as follows:

Given a time-stamped set of prioritized wildfire tasks, a Walker observer constellation, a ground-station configuration, finite communication opportunities, and a configurable fraction of central nodes, determine how architecture and centralization affect task reception, dissemination breadth, follow-up observation, communication burden, and decision robustness.

The phrase *decision robustness* is used for stability under objective weights. Physical robustness refers separately to the retention of mission performance after injected node failures. Keeping those

two meanings separate prevents a stable ranking from being mistaken for a failure-tolerant architecture.

1.3 Aim and Objectives

The aim is to establish a defensible performance–resource–robustness trade space for wildfire task allocation in a distributed satellite system. The objectives are:

1. construct a traceable wildfire-task dataset from Sentinel-3 SLSTR FRP products, supported by FIRMS screening and event validation;
2. adapt the task-allocation framework to multiple release times, priority classes, task sizes, and deadlines;
3. model finite, time-dependent communication windows and evaluate observation only after task reception;
4. execute and validate the complete 891-case Walker and central-node design grid for two event regions and the expanded 539-scenario Fresh V2 campaign where available;
5. quantify central-node, orbit, constellation-size, communication-load, and regional effects without assigning causal meaning to unadjusted correlations;
6. identify exact Pareto sets and evaluate the stability of architecture rankings under alternative objective weights; and
7. quantify robustness through explicit random and targeted node-removal, link-window, ground-station, and capacity-degradation experiments rather than through nominal failure labels.

Table 1.1 maps each objective to the evidence used in this draft. The two-region nominal layer is complete. Fresh V2 robustness, task-size, policy, and revised weight analyses are complete for California; the corresponding India Fresh V2 extension is not locally available and is not inferred.

Table 1.1 Objectives, evidence products, and status in this working draft.

No.	Objective	Primary evidence	Status
O1	Sentinel-3 wildfire task construction	De-duplicated 805-task CSV and processing history	Complete
O2	Multi-task priority and deadline model	Case-level <code>wildfire_tasks_used.csv</code>	Complete
O3	Capacity-aware dissemination and observation-after-reception	Windows, transfers, receptions, and observation results	Complete
O4	Full architecture campaign	1,782 earlier nominal cases; California Fresh V2 scenario archive	Nominal complete; Fresh V2 California complete
O5	Architecture and regional effects	Earlier paired-region tables and Fresh V2 California factor effects	Partial for final V2 regional comparison
O6	Pareto and weight sensitivity	Exact Pareto membership and 10,000 revised weight draws	Complete for Fresh V2 California
O7	Failure-injection robustness	Random/targeted failures, outages, and capacity degradation	Complete for Fresh V2 California

1.4 Research Questions

The approved questions are made operational through the exported metrics:

- RQ1. How can a PBL-focused distributed task-allocation framework be adapted to generate and process multiple time-critical wildfire tasks from real remote-sensing detections?
- RQ2. How do real wildfire task distributions, confidence values, fire intensity, and regional task density affect dissemination and observation fulfilment?
- RQ3. How does the fraction of central nodes influence first-reception latency, full dissemination, deadline success, observation fulfilment, routing depth, communication load, and architecture dependency?
- RQ4. Which Walker constellation configurations provide the best trade-off among performance, complexity, and robustness for wildfire monitoring?
- RQ5. How stable is the preferred central-node fraction under changes in task size, observation radius, failure mode, and objective-function weights?

The available evidence answers RQ1 and the California-specific portions of RQ2–RQ5. Fresh V2 supplies validated California policy, task-size, failure, and weight experiments. A final Fresh V2 answer about regional load/generalization requires the India archive; observation-radius sensitivity and a separately extracted first-reception latency also remain open.

1.5 Hypotheses

Five hypotheses organize the analysis:

H1: Diminishing returns

Increasing central-node fraction initially reduces dissemination latency and improves task fulfilment, but the marginal benefit diminishes after the network becomes sufficiently connected.

H2: Resource penalty

Central-node count, communication opportunity, and dependency increase with centralization, so the fastest architecture need not be the preferred architecture.

H3: Load dependence

A dense task region produces different capacity and ranking behaviour from a sparse event region even when the architecture grid is identical.

H4: Geometry dependence

Successful reception is necessary but not sufficient for observation because a receiving satellite still needs a valid access before the deadline.

H5: No universal optimum

The preferred case changes with region and objective weights; a defensible result is a Pareto set or stable candidate set rather than a universal scalar optimum.

1.6 Scope and Boundaries

Table 1.2 defines the scope hierarchy. Central-node fraction is the primary variable. Walker geometry supports the centrality analysis, while task size, observation radius, and node failure test the sensitivity

Table 1.2 Hierarchy of the implemented and planned thesis scope.

Level	Study element	Treatment in this report
Core	Central-node fraction in a wildfire DSS	Complete 0–100% sweep for every base architecture
Supporting	Satellite count, plane count, altitude, inclination	Full-factorial nominal screening
Sensitivity	Objective weights	Complete 10,000-draw Fresh V2 California analysis
Sensitivity	Task size and injected failures	Complete for Fresh V2 California
Sensitivity	Observation radius	Fixed at 700 km; targeted sweep not executed
Extension	Dedicated heterogeneous central-node orbit	Conceptual discussion and future work only
Out of scope	Onboard fire-detection latency, weather, cloud, pointing, energy, and monetary cost	Explicitly excluded from final claims

of the main conclusion. A separate heterogeneous relay layer is outside the implemented nominal campaign.

The simulated mission chain starts when a processed wildfire task is injected. It does not simulate the original thermal measurement, onboard detection, cloud screening, product generation latency, or delivery of the final alert to civil-protection authorities. The observation radius is a mission-level access approximation and not a detailed optical-instrument pointing model. “Cost” means a declared non-monetary resource proxy; no procurement or lifecycle cost is inferred.

1.7 Meaning of a Central Node

The conceptual system definition and the implemented experiment are intentionally separated. Conceptually, central nodes may differ from ordinary satellites in hardware, processing, communication links, routing, coordination, and connection to ground. The final simulator represents only the portion for which validated code and outputs exist: selected nodes receive a central-node label, communication opportunities follow the configured link policy, and the routing policy preferentially forwards through the central-node structure. The campaign does not assign different sensor performance, processor speed, storage volume, reliability, or monetary cost to central nodes.

This distinction is essential for interpretation. A measured central-node effect is an effect of the implemented communication and routing role under common spacecraft geometry. It is not evidence that a particular hardware implementation is optimal. Hardware and operations can later be attached to the architectural role once real subsystem assumptions are available.

1.8 Scientific and Engineering Contributions

The thesis contributions are:

1. a real-data, de-duplicated Sentinel-3 task set that connects FRP, confidence, spatial clustering, priority, data size, and response deadline;
2. a modular DSS workflow separating physical communication and observation opportunities from the allocation/evaluation layer;

3. a capacity-aware and time-ordered forwarding evaluation that records task reception, transfer, observation, and failure evidence;
4. a complete two-region, 891-case-per-region nominal campaign plus a California Fresh V2 campaign with operational, policy, task-size, failure, and combined-stress scenarios;
5. a clear distinction between observation fulfilment, conditional reception latency, receiver coverage, and useful-recipient full dissemination;
6. exact Pareto filtering and Monte Carlo objective-weight sensitivity without claiming a universal optimum; and
7. reproducible random and targeted failure experiments with recorded seeds, separate original/-surviving denominators, and time-aware mission metrics.

The author developed the wildfire pipeline, integration, simulation automation, campaign execution, post-processing, statistical analysis, and report. Vincenzo Messina provided the communication model and the TAT-C basis. This allocation is stated here because the software components directly shape the interpretation of the results.

1.9 Report Structure

Chapter 2 presents the literature review. Chapter 3 explains the Sentinel-3 task dataset and deduplication. Chapter 4 formalizes the Fresh V2 architecture, communication, routing, observation, and metrics. Chapter 5 separates the earlier two-region nominal evidence from the expanded Fresh V2 scenario design and validation. Chapter 6 reports the earlier validated regional baseline. Chapter 7 reports the available Fresh V2 California operational, policy, task-size, robustness, Pareto, and weight results. Chapter 8 answers the research questions with explicit evidence boundaries, while Chapter 9 gives the conclusions and remaining work. The appendices provide metric definitions, campaign configuration, validation records, supplementary tables, and software-release requirements.

2 Background and Related Work

2.1 Distributed Satellite Systems

A distributed satellite system (DSS) consists of multiple space assets that cooperate to achieve mission objectives that would otherwise be executed by an individual spacecraft or a centrally planned constellation. Distribution can occur across sensing, processing, communication, decision-making, and operations. The design space therefore spans tightly coordinated constellations, federated systems that share resources across mission owners, and autonomous networks that allocate observations onboard [Golkar and Cruz, 2015, Scrocciolani et al., 2024]. These categories should not be treated as interchangeable. A constellation describes an orbital and mission organization, whereas federation and decentralization describe how capabilities, information, and authority are shared.

Automation becomes increasingly important as the number of assets and requests grows. Reviews of large DSS operations identify a shift away from spacecraft-by-spacecraft commanding toward goal-based planning, automated monitoring, and coordination mechanisms that scale with the system rather than with the number of human operators [Ben-Larbi et al., 2021]. This operational perspective is directly relevant to time-critical Earth observation: a constellation may have adequate geometric coverage, yet still respond too slowly if every new request must travel through a remote planning loop before any spacecraft can act.

Centralized coordination has two main advantages: global information can be combined into a coherent decision, and policy enforcement is comparatively simple. Its weaknesses are latency to the decision point, dependence on ground contact or a coordinating node, and scaling difficulty as task and asset counts increase. Fully decentralized coordination can improve local responsiveness and remove a single decision bottleneck, but may require repeated message exchange and may produce inconsistent local views under intermittent connectivity. Recent work consequently motivates decentralized satellite networking and onboard decision-making, while also recognizing the resulting coordination and protocol challenges [Oh and Vasisht, 2024].

Hybrid architectures occupy the space between these endpoints. Selected satellites can be assigned enhanced routing or coordination roles without making one node the permanent controller of the mission. This introduces an architectural design variable: how many enhanced nodes are needed, where they are placed, and what privileges they possess. A hybrid architecture is useful only when those privileges are explicit. Calling a satellite a “central node” does not itself improve performance; the effect must arise through defined link access, forwarding eligibility, processing capacity, storage, or coordination rules.

2.2 Task Allocation and Observation Scheduling

Earth-observation scheduling is a constrained resource-allocation problem involving target visibility, time windows, instrument constraints, task value, spacecraft resources, and communication opportunities [Ferrari et al., 2025]. A planner must decide not only which requests are valuable but also whether a feasible sequence of observations exists. In a dynamic mission, new requests can arrive after a schedule has already been constructed, so the system must repeatedly trade schedule stability against responsiveness.

Large constellations make a single global optimization increasingly difficult to compute and communicate. Decentralized approaches partition the allocation problem across satellites, allowing agents to exchange local requests, bids, or partial schedules [Parjan and Chien, 2023, Zilberstein et al., 2025].

Such methods can reduce dependence on a single planner and permit decisions nearer to the sensing assets. Their benefit, however, depends on the communication model: an allocation algorithm that assumes instantaneous exchange can look effective even when the corresponding messages could not reach the agents before the observation deadline.

It is therefore essential to distinguish decision quality from information availability. A scheduled observation and a disseminated task are different system states:

Task generation	A wildfire event is converted into a request containing location, creation time, priority, deadline, and data-size assumptions.
Task reception	A satellite has received sufficient task information to act on the request.
Observation opportunity	Satellite geometry permits sensing the task location within the assumed field of regard or observation radius.
Observation fulfilment	An opportunity occurs after reception and before the task deadline, and the observation is assigned or executed under the model.
Data delivery	The resulting observation product is transferred to a ground station or another defined endpoint.

The separation exposes two different failure modes. A task can miss its deadline because no satellite has a suitable observation opportunity, even though dissemination was immediate. Conversely, a geometrically feasible opportunity can be lost because the task reached the observing satellite too late. Reporting only final observation success hides this distinction and makes it difficult to tell whether an architectural change improved coverage, communication, or both. For that reason, this thesis treats first reception, dissemination extent, observation fulfilment, and latency as separate metrics.

The present work focuses on dissemination of post-injection wildfire tasks and the subsequent observation opportunity. It does not simulate the complete chain from onboard thermal-anomaly detection through image-product downlink. Downlink latency studied in early baseline work remains useful context, but it cannot substitute for communication-aware task-reception metrics unless the mission boundary is explicitly extended.

2.3 Time-Varying Satellite Connectivity

Graph-based representations separate physical contact opportunities from the communication process. Let the time-dependent satellite network be

$$G(t) = (V, E(t)), \quad (2.1)$$

where V contains observer satellites, enhanced central nodes, and optionally ground stations, while $E(t)$ contains links available at time t . A contact window is a maximal interval over which a specific edge exists. A transfer event is an actual use of that edge to transmit a task. The distinction is important: a system may generate many windows but use only a small fraction of them, or repeatedly use a small number of bottleneck links.

A static snapshot cannot fully characterize such a network. Two nodes that are never connected by an instantaneous end-to-end path may still communicate over a temporal path if intermediate satellites store the task and forward it during later contacts. Delay- and disruption-tolerant networking formalizes this store–carry–forward behavior. Bundle Protocol version 7 is designed for stressed networks with intermittent connectivity and variable delay [Burleigh et al., 2022]. For environments in which future contacts are predictable, Contact Graph Routing and the CCSDS Schedule-Aware Bundle Routing standard compute routes from a time-ordered contact plan rather than from a permanently connected topology [Burleigh, 2008, Araniti et al., 2015, Consultative Committee for Space Data Systems, 2019].

For a contact $e = (i, j, t_s, t_e, r)$, the available data volume is bounded by

$$C_e = r(t_e - t_s), \quad (2.2)$$

where r is the assumed link rate and $[t_s, t_e]$ is the contact interval. A feasible dissemination path must therefore respect contact order, residual capacity, task creation time, and task deadline. Hop count alone is an incomplete proxy: a two-hop temporal path can be slower than a longer path if the second contact occurs much later. Similarly, unlimited copying does not guarantee timely completion when contacts are unavailable or their capacity is exhausted.

Useful structural descriptors include degree, density, connected components, average path length, betweenness centrality, algebraic connectivity, graph energy, and concentration of transfer traffic. In a temporal communication study these descriptors are explanatory variables, not mission outcomes. They can suggest why dissemination changes—for example, shorter temporal paths or a concentration of traffic on a few nodes—but deadline success must still be recomputed from timestamped transfers. This is also why an instantaneous adjacency matrix cannot by itself establish robustness: resilience depends on when a failed node or link was needed, what alternative future contacts existed, and whether remaining storage and capacity were sufficient.

2.4 Central Nodes as a Hybrid Architectural Mechanism

The literature often contrasts a central planner with fully decentralized agents, but a routing-enhanced central node is a more limited mechanism. It can act as a preferred relay, an information broker, or an interface to a restricted communication domain while observation authority remains distributed. This distinction matters because three forms of centralization can otherwise be conflated:

- *decision centralization*, in which one agent chooses allocations for the system;
- *information centralization*, in which one or more nodes maintain a broader state estimate; and
- *communication centralization*, in which forwarding is restricted or preferentially routed through selected nodes.

These forms have different benefits and failure dependencies. Adding relay-eligible nodes may reduce the waiting time to a usable path, yet it may also increase duplicate traffic or concentrate load. A high central-node fraction can therefore exhibit diminishing returns, and in a restrictive policy it can even reduce the number of useful forwarding relationships. Conversely, a small fraction may be valuable if the selected nodes bridge otherwise separated temporal components. The architectural question is not simply whether centralization is “good” or “bad”, but how performance changes as enhanced routing participation is varied under one declared policy.

Network analysis of federated satellite systems supports studying this question across both mission and graph metrics [Scrocciolani et al., 2024]. However, correlations between a structural metric and performance do not prove that the structural change caused the outcome. Satellite count, orbital planes, altitude, inclination, and central-node fraction all alter multiple aspects of geometry and load. A defensible trade study therefore combines controlled factor sweeps, architecture-paired comparisons, and explicit resource measures with the network descriptors.

2.5 Wildfire Detection from Space

Satellite active-fire products identify thermal anomalies and report attributes such as acquisition time, location, confidence, and fire radiative power (FRP). FRP expresses the instantaneous rate of radiant energy emitted by a fire and is commonly reported in megawatts. It provides an intensity-related variable that is more informative for task ranking than a detection count alone, although retrieval uncertainty and sensor sampling must be retained when interpreting it [Wooster et al., 2005].

NASA FIRMS distributes active-fire detections from instruments including the Visible Infrared Imaging Radiometer Suite (VIIRS). The VIIRS 375 m product uses the higher spatial resolution imagery bands and was developed to improve the detection and mapping of small fires relative to coarser products [Schroeder et al., 2014]. FIRMS is therefore suitable for rapid event screening and for constructing a spatially and temporally resolved task load. It does not, however, provide a list of statistically independent fires: adjacent pixels and repeated overpasses can describe the same evolving event.

Sentinel-3's Sea and Land Surface Temperature Radiometer (SLSTR) provides an independent active-fire and FRP product. Its fire-oriented channels and algorithms were designed and evaluated before launch [Wooster et al., 2012], and the operational daytime product was subsequently assessed against MODIS, VIIRS, and Landsat active-fire data [Xu and Wooster, 2023]. The different spatial sampling, overpass times, spectral response, cloud screening, and confidence conventions mean that SLSTR and VIIRS detections should not be merged row by row. Agreement or disagreement can arise from the observation process as well as from real fire evolution.

2.6 From Fire Pixels to Communication Tasks

Turning remote-sensing detections into simulation requests requires an explicit abstraction layer. First, detections must be filtered to a declared event region and acquisition interval. Second, spatially adjacent or temporally repeated pixels must be clustered or deduplicated so that the simulator does not treat every pixel as an unrelated emergency. Third, a priority, deadline, and message size must be assigned using reproducible rules. Finally, the generated task table must retain provenance to its source product and extraction time.

This transformation is scientifically consequential. A denser task list increases contention for communication windows and observation opportunities; a shorter deadline increases censoring even if the same contacts exist; a larger message consumes more capacity at each forwarding hop. Priority can also become confounded with task size or deadline if all three are derived from FRP. Sensitivity to message size is therefore best measured by applying controlled multipliers to an otherwise fixed task set, rather than by comparing native priority classes and attributing every difference to size.

The two datasets consequently serve different but compatible roles in this thesis:

Table 2.1 Division of roles between FIRMS VIIRS and Sentinel-3 SLSTR.

Aspect	FIRMS VIIRS	Sentinel-3 SLSTR FRP
Primary role	Broad and rapid event screening; task-density construction; near-real-time-compatible extraction.	Independent product comparison; validation of intensity and confidence behavior; sensor-specific case studies.
Native confidence	Categorical values such as low, nominal, and high.	Product-specific numerical confidence fields in the processed dataset.
Strength	Fine active-fire detail and convenient access through FIRMS services.	Dedicated FRP product from an independent observing system.
Risk	Repeated pixels or widespread low-value burning can inflate the apparent task count unless clustered and filtered.	Different sampling and product conventions prevent direct one-to-one comparison without temporal and spatial matching.

The final experiment must state which product generated each simulation task list. Combining products without a matching, deduplication, and provenance rule would risk double counting. A de-

fensible approach is to use one source for the operational task generation and the other for validation or sensitivity analysis.

2.7 TAT-C and Supporting Orbital Tools

The Trade-space Analysis Tool for Constellations (TAT-C) provides mission-level functions for defining spacecraft, instruments, coverage grids, observation access, revisit, and latency [TAT-C Development Team, 2026]. In the early project stages, Skyfield was used as an independent orbit and line-of-sight tool, while TAT-C was used for mission-level access and aggregation. The later DSS framework places a custom allocation and communication layer above the orbit and access calculations.

This separation establishes a traceable simulation chain:

- the orbital layer determines where satellites and ground stations are and when geometric access exists;
- the communication layer converts geometry into time-bounded link capacity;
- the allocation layer selects which task is transmitted over which link;
- the observation layer checks whether a receiving satellite can subsequently observe the task; and
- the post-processing layer computes completion, latency, cost proxies, and trade-space rankings.

It also prevents one model component from silently compensating for another. For example, widening the observation radius may improve observation success but cannot be interpreted as a networking improvement. Likewise, adding forwarding copies can improve reception but does not change orbital access. Layer-specific metrics are needed before an end-to-end outcome can be explained.

2.8 Synthesis of Related Work and Thesis Positioning

Table 2.2 summarizes the literature streams that define the thesis boundary. The relevant gap is their combination, not the absence of research in any one stream.

Table 2.2 Related-work synthesis and positioning of the thesis.

Literature stream	Primary contribution	Boundary relative to this thesis
Federated and distributed satellite systems [Golkar and Cruz, 2015, Scrocciolani et al., 2024, Ben-Larbi et al., 2021]	Resource sharing, system organization, network descriptors, and scalable operations.	Does not by itself quantify deadline-aware wildfire-task propagation across a controlled central-node sweep.
Decentralized observation allocation [Parjan and Chien, 2023, Zilberstein et al., 2025]	Distributed algorithms for assigning observation requests in large constellations.	Allocation quality must be coupled to a finite-capacity, time-varying communication model to establish when agents actually know a task.
Space DTN and schedule-aware routing [Burleigh et al., 2022, Araniti et al., 2015, Consultative Committee for Space Data Systems, 2019]	Store–carry–forward communication and routing over scheduled contacts.	Provides the networking concepts, but not the wildfire task abstraction or architecture-level observation trade study.
Satellite active-fire products [Schroeder et al., 2014, Wooster et al., 2012, Xu and Wooster, 2023]	Detection, confidence, and FRP retrieval from VIIRS and SLSTR.	Sensor detections must be converted into deduplicated requests with explicit deadlines, priorities, and data volumes.
This thesis	Communication-aware wildfire-task dissemination followed by observation, evaluated over a satellite architecture grid and central-node fractions.	Claims are limited to the implemented routing policy, task sets, contact model, and resource proxies; peer-to-peer policy, failure injection, and controlled task-size sweeps require separate experiments.

2.9 Research Gap

Existing work establishes the value of federated satellite resources, decentralized observation scheduling, schedule-aware routing, and satellite active-fire products. Fewer studies connect all four levels in a reproducible architecture experiment. The gap targeted here is the joint analysis of (i) real, heterogeneous wildfire-derived task loads, (ii) a variable fraction of routing-enhanced central nodes, (iii) finite contact duration and message size, (iv) observation only after successful task reception, and (v) architecture performance evaluated together with communication burden and dependency cost.

The scope is deliberately narrower than general autonomous constellation scheduling. The thesis evaluates one implemented communication policy across a declared architecture grid and two event-specific task loads. This supports claims about architectural trade-offs within that model, including conditional latency, dissemination completion, and rank sensitivity. It does not turn a zero-central-node case into a peer-to-peer policy comparison, and nominal outcome failures are not equivalent to robustness under injected node or link outages. Stating these boundaries converts the literature gap into a testable contribution rather than a claim of universal optimality.

3 Wildfire Task Dataset and Preprocessing

3.1 Role of the Wildfire Dataset

The task dataset is the interface between fire remote sensing and the DSS experiment. The simulation does not ingest raw radiance. It receives a task table in which each row represents a fire region that may justify a follow-up observation. The data pipeline must therefore reduce repeated pixels and repeated overpasses without erasing physically distinct events, retain traceability to the source products, and transform heterogeneous sensor confidence into an operational ranking variable.

The complementary roles of FIRMS screening and Sentinel-3 task generation are summarized in Table 2.1.

FIRMS was used during development for rapid event-day screening and independent support. Sentinel-3 SLSTR Level-2 FRP products were selected as the primary task source. This separation uses the strengths of each source: FIRMS provides convenient, dense active-fire screening, while Sentinel-3 provides the mission-specific product used to create the final tasks [[NASA Fire Information for Resource Management System, 2026](#), [European Space Agency, 2026](#), [Schroeder et al., 2014](#), [Wooster et al., 2012](#), [Xu and Wooster, 2023](#)].

3.2 Event Windows and Regional Roles

Two three-day event windows were selected before the architecture campaign. The California window covers 3–5 July 2025 and contains the early activity of the Madre Fire. The India window covers 14–16 April 2026 and intersects reported forest-fire activity in the Bihar and Uttar Pradesh region. Both regions are retained because they create complementary experimental conditions.

California is a compact event-validation case. It contains 32 final tasks, including the only Critical task in the dataset and six High-priority tasks. India contains 773 tasks distributed over a much larger area and is therefore a high-load stress case. The regional label consequently represents a combination of geography, event selection, task density, priority mix, and release times. A regional difference must not be interpreted as a pure geographic effect.

Figure 3.1 will show the centroids and priorities of the final tasks. The strong imbalance in point density is not a plotting artifact: it is the workload contrast used in the experiment.

3.3 Processing Pipeline

Figure 3.2 summarizes the task-generation workflow. Product discovery and NetCDF extraction are followed by filtering, clustering, scoring, de-duplication, and export. Every transformation has a narrower purpose than the previous one: filtering rejects weak or irrelevant detections, clustering forms candidate fire regions, scoring ranks those regions, and de-duplication determines which candidates become independent DSS tasks.

3.3.1 Product and NetCDF selection

A downloaded .SEN3 directory corresponds to one SLSTR Level-2 FRP product granule. The product name contains the sensing interval, while a later timestamp records product generation. The region

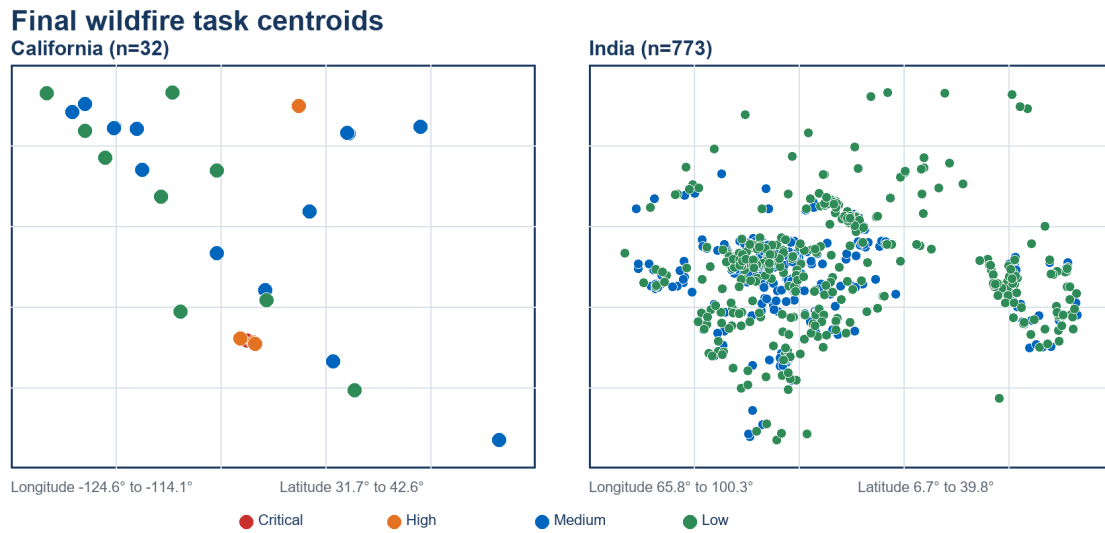


Figure 3.1 Final de-duplicated Sentinel-3 wildfire-task centroids for California and India, coloured by priority class. The figure is generated directly from the 805-row task input.

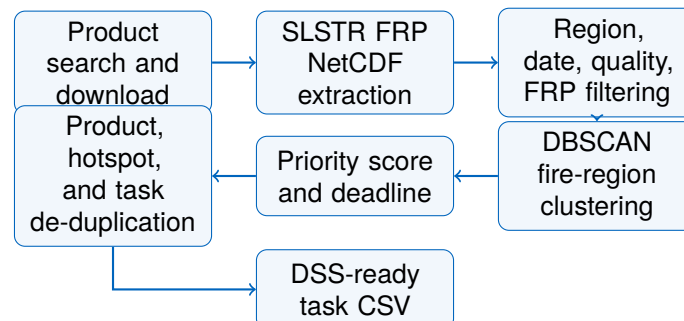


Figure 3.2 Sentinel-3 wildfire-task generation and export workflow.

of interest is applied after reading the geolocated fire pixels because the product is not intrinsically limited to California or India.

The first pipeline version successfully automated search, download, extraction, and scoring but read repeated product content. The second version corrected invalid confidence values and exact duplicates. The third version, used for the final task input, introduced four de-duplication levels:

1. **product level:** each .SEN3 product is read once;
2. **NetCDF level:** the primary FRP file is selected instead of indiscriminately reading support files;
3. **hotspot level:** exact and near-duplicate detections are removed using rounded position, time, and FRP; and
4. **task level:** nearby candidates representing the same fire region on the same day are merged, retaining the highest-priority representative.

The fall in regional total FRP between versions is therefore an expected consequence of removing repeated observations, not a loss of physical signal. Table 3.1 records this development history. The values are included to document why the third version is used and are not architecture results.

Table 3.1 Sentinel-3 pipeline refinement and regional total FRP reported during development.

Version	Main change	California FRP (MW)	India FRP (MW)
v1	Initial automated search, extraction, and scoring; repeated reading remained	137,753	323,677
v2	Revised confidence handling and exact hotspot de-duplication	68,876	161,811
v3	Product-, NetCDF-, hotspot-, and task-level de-duplication	22,686	34,595

3.3.2 FRP threshold and clustering

Fire radiative power is used as an intensity variable because it is related to active biomass combustion and is more informative than an unweighted detection count [Wooster et al., 2005]. Development experiments applied an FRP threshold of 10.00 MW before clustering. This threshold removes weak thermal detections from task generation while retaining operationally meaningful active-fire signals.

DBSCAN groups detections using spatial density rather than a pre-declared number of clusters. Radius and minimum-sample sweeps were used during development to avoid the two extremes of treating every pixel as a separate task or merging geographically distant fires. A 10 km radius and `min_samples=3` were used in the documented FIRMS screening work; the Sentinel-3 task output retains the cluster centroid, approximate radius, number of detections, and product provenance. Isolated detections are preserved or rejected according to the pipeline rules rather than silently folded into the nearest cluster.

3.4 Confidence Harmonization

FIRMS and Sentinel-3 expose confidence differently. Sentinel-3 provides a numerical MWIR confidence percentage, which is divided by 100. FIRMS VIIRS provides categorical values that were mapped to 0.33, 0.66, and 1.00 for low, nominal, and high confidence. The harmonized value is a relative ranking variable, not a calibrated probability that a detection is a wildfire.

For Sentinel-3, invalid values such as -1 are not interpreted as zero confidence. They receive a neutral score of 0.50 and are flagged as missing. This prevents an unavailable field from suppressing a

task while retaining transparency for later filtering. The task table also records the missing-confidence fraction and the minimum, maximum, and dominant confidence information of the contributing detections.

3.5 Priority Score

The task-priority score combines fire intensity, confidence, and cluster size. The FRP component first combines normalized total and maximum FRP:

$$S_{\text{FRP}} = 0.60\mathcal{N}(\log(1 + \text{FRP}_{\text{total}})) + 0.40\mathcal{N}(\log(1 + \text{FRP}_{\text{max}})), \quad (3.1)$$

where $\mathcal{N}(\cdot)$ denotes the declared normalization within the scoring dataset. The logarithm prevents a small number of extreme fires from determining the entire range. The final score is

$$S_{\text{task}} = 0.50S_{\text{FRP}} + 0.30S_{\text{conf}} + 0.20S_{\text{cluster}}. \quad (3.2)$$

Equation 3.2 gives FRP the largest weight because it represents energetic strength. Confidence acts as a reliability modifier, and cluster size represents local spatial extent or density. The weights are an engineering model, not a learned clinical or civil-protection severity scale. This limitation is important because agricultural burning and industrial heat sources may still receive a high score if their physical inputs are large.

3.6 Final Task Population

Table 3.2 describes the final task population. The India event contributes 96.0% of the tasks, but California contains a greater fraction of urgent tasks. This is why regional totals are normalized per task before communication burden is compared.

Table 3.2 Final de-duplicated Sentinel-3 task population used by the nominal campaign.

Region	Critical	High	Medium	Low	Total
California	1	6	16	9	32
India	0	11	395	367	773
Total	1	17	411	376	805

Figure 3.3 presents both absolute counts and within-region shares. The left panel makes the communication-load difference explicit, while the right panel shows that California has a markedly larger urgent-task share. Reporting both views avoids letting the India count dominate the visual interpretation.

Priority determines both the response deadline and nominal packet size as shown in Table 3.3. These packet sizes represent the task-allocation message carried by the simulated network, not raw imagery. Therefore, the nominal communication results concern dissemination of observation requests rather than transfer of full SLSTR products.

Because priority changes simultaneously with size and deadline, the nominal campaign is not a controlled task-size experiment. A comparison between Critical and Low packets would mix data volume, urgency, geography, and fire severity. Chapter 7 therefore defines a separate task-size sweep that holds task identity and deadline constant while applying controlled multipliers.

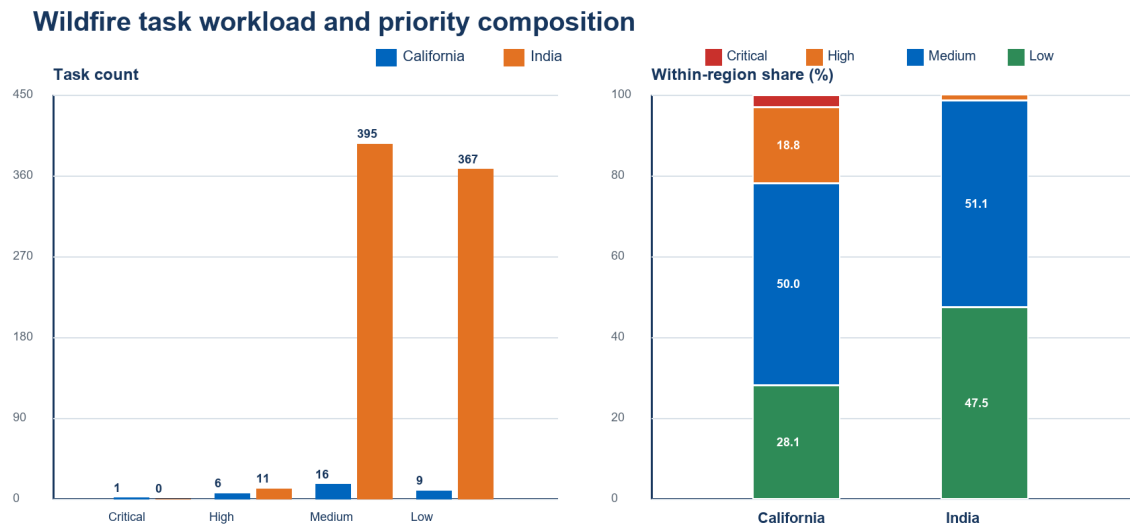


Figure 3.3 Absolute regional workload and priority composition of the final task set.

Table 3.3 Priority-dependent response deadlines and task packet sizes in the final nominal campaign.

Priority	Deadline (min)	Packet size (Mbit)	Operational interpretation
Critical	30	2.0	Immediate high-severity follow-up request
High	60	1.5	Urgent follow-up request
Medium	180	1.0	Time-sensitive routine request
Low	360	0.5	Lower-urgency monitoring request

3.7 DSS Task Schema

The source and simulator fields are summarized in Table 3.4. Every field shown is either present in the de-duplicated CSV or created deterministically by the task loader. The complete schema is retained in the appendix and software release.

Table 3.4 Core task fields transferred from the Sentinel-3 pipeline to the DSS simulator.

Field	Unit/type	Meaning
task_id	string	Stable task identifier within the source table
region, date	category/date	Event-region membership and source day
timestamp_utc	UTC time	Task release time before deadline construction
centroid_lat/lon	degrees	Representative fire-region position
num_detections	count	Detections supporting the clustered task
approx_radius_km	km	Approximate spatial radius of the task cluster
mean/max/total_frp_ mw	MW	Fire-radiative-power summaries
confidence_score	0–1	Harmonized relative confidence score
task_priority_score	0–1	Composite score from Equation 3.2
priority_class	category	Critical, High, Medium, or Low
required_response_ time_min	min	Deadline offset from task release
task_size_mbits	Mbit	Simulator-assigned packet size

3.8 Validation and Limitations of the Task Dataset

The strongest event-level validation is the Madre Fire. Both Sentinel-3 and FIRMS produced high-priority tasks near the reported event location during the California window. The India task set has broader regional support but weaker named-incident traceability. The data are consequently appropriate for architecture stress testing and selected event validation, but they are not a labelled benchmark for wildfire classification accuracy.

Several limitations remain. Bounding-box filtering can retain points outside the political region unless a polygon mask is applied. Cloud, smoke, and overpass time change the detections available to each sensor. FIRMS and Sentinel-3 cannot be compared by regional FRP alone because their sensing, sampling, and de-duplication differ. A more defensible agreement study would match tasks in space and time and report centroid distance, priority-class agreement, FRP ratio, and unmatched counts.

For this thesis, the final task table is frozen before architecture evaluation. All 891 cases within a region use the same task source and row count, which prevents architecture comparisons from being contaminated by changing wildfire inputs. The remaining uncertainty belongs to external validity: the preferred architecture could differ for another season, sensor, region, or priority model.

4 System Model and Methodology

4.1 Methodological Overview

The final workflow converts frozen wildfire detections into time-dependent tasks, evaluates a factorial Walker-constellation design space, and replays each architecture under nominal, policy, load, and failure scenarios. The workflow deliberately separates four layers: task data, physical opportunities, allocation logic, and evaluation. This separation makes it possible to change a routing policy or inject a failure without silently changing the task set or orbital geometry.

Figure 4.1 shows the dependency chain. A task cannot be sent before release, a transfer cannot exceed a contact-window capacity, and an observation is successful only if an observing satellite already holds the task and acts before the deadline.

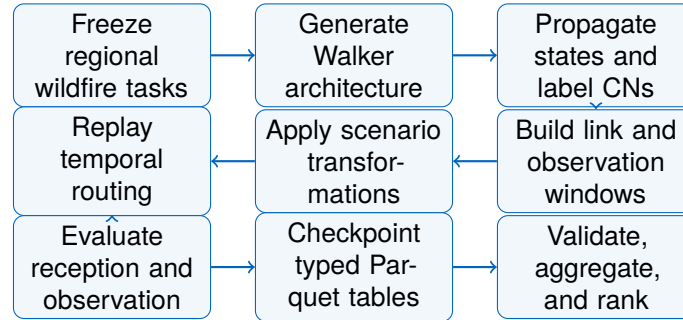


Figure 4.1 Fresh Full891 V2 simulation and post-processing workflow.

4.2 Task Input and Temporal Horizon

The campaign consumes a deterministic CSV rather than querying a live service during simulation. The file contains 805 frozen tasks: 32 for California and 773 for India. Each row supplies a stable identifier, region, position, release epoch, priority, deadline, task-message size, and source provenance. Chapter 3 describes the upstream Sentinel-3 processing and the supporting FIRMS screening.

The original UTC epochs are retained. For each region, the simulation duration is

$$T_{\text{sim}} = \max(72 \text{ h}, d_{\text{max}} - r_{\text{min}} + 6 \text{ h}), \quad (4.1)$$

where r_{min} is the first release and d_{max} is the latest deadline. This gives approximately 73.134 hours for California and 73.659 hours for India in Fresh V2. Priority-dependent deadlines are 30, 60, 180, and 360 minutes for Critical, High, Medium, and Low tasks. The associated nominal messages are 2.0, 1.5, 1.0, and 0.5 Mbit. These are simulated task descriptions, not downloaded sensor-image volumes.

4.3 Walker Constellation and Central-Node Assignment

An architecture is represented by total satellite count T , plane count P , altitude H , inclination I , central-node fraction C , and phasing $F = 1$. The complete grid is

$$\{60, 120, 180\} \times \{3, 5, 10\} \times \{500, 800, 1000\} \times \{60, 80, 98.6\} \times \{0, 10, \dots, 100\}, \quad (4.2)$$

which gives $3^4 \times 11 = 891$ cases per region. For example,

$$\text{T060_P10_H0800_I0986_CN040_F1} \quad (4.3)$$

denotes 60 spacecraft in 10 planes at 800.00 km, 98.60° inclination, with 40 percent (24) assigned the central-node role.

The nodes share the same orbital and sensing model; “central” denotes a communication and routing role, not a separate relay shell or a fully costed hardware class. Central nodes are spread across planes as evenly as integer constraints permit and then spaced within each plane. This deterministic placement prevents arbitrary clustering from confounding a fraction sweep. It is a reproducible baseline rather than a placement optimum.

4.4 Orbit and Observation Opportunity Model

The orbital layer uses circular two-body propagation on a spherical rotating Earth, consistent with the integrated TAT-C-based workflow [TAT-C Development Team, 2026]. Fresh V2 uses $R_E = 6378.14$ km, $\mu = 398\,600.44 \text{ km}^3 \text{ s}^{-2}$, Earth rotation 7.2921159×10^{-5} rad/s, and a 60.00 s time step. Walker plane population and phasing are validated for every generated case.

An observation opportunity exists when the satellite–task ground distance is within the fixed 700.00 km access radius. Let O_{is} be candidate times for task i and satellite s , r_i the release time, d_i the deadline, and a_{is} the time at which s receives the task. A valid follow-up time is

$$t_i^{\text{obs}} = \min_s \{t \in O_{is} : t \geq \max(r_i, a_{is}) \wedge t \leq d_i\}. \quad (4.4)$$

The access radius is a mission-level approximation. It does not model cloud, smoke, lighting, field of view, slew, image quality, energy, storage, or conflicts between simultaneous observations. No Fresh V2 observation-radius sweep is presently available.

4.5 Communication Opportunities and Link Capacity

All unordered satellite pairs are first tested for physical contact; scenario topology rules are applied later. The ground segment contains the Munich station with a 10.00° minimum elevation. Each communication window stores endpoints, type, start and end time, effective data rate, usable duration, and capacity.

Satellite-to-satellite capacity is calculated by the active V19 UHF link-budget implementation, using a 2.00 W, 437.00 MHz, 9.60 kHz channel and 9.6 ksymbol/s signal together with the configured gains, losses, sensitivity, and a 1.5 central-node link-budget boost. The resulting effective rate is capped near $19.20 \text{ kbit s}^{-1}$. A legacy configuration field named `sat_sat_data_rate_bps` contains 250,000 bit/s but is not consumed by the active calculation and is therefore not used as a scientific claim. Satellite-to-ground windows use $100.00 \text{ kbit s}^{-1}$.

For window w of usable duration Δt_w , utilization factor η_w , and effective rate R_w , the capacity is

$$B_w = \frac{R_w \Delta t_w \eta_w}{10^6} \quad [\text{Mbit}]. \quad (4.5)$$

Fresh V2 sets $\eta_w = 1$. A window records potential capacity; a transfer event records actual use. These quantities are never treated as interchangeable.

4.6 Topology and Temporal Routing

Three topology-policy variants are replayed:

- Central-only** An inter-satellite link is usable only when at least one endpoint is central. CNO therefore has no inter-satellite link and relies on ground uplink; it is not a peer-to-peer baseline.
- Peer-only** All feasible inter-satellite links are usable and routing does not privilege the central label.
- Hybrid** All feasible links are usable and the selection order prefers central-node forwarding.

Ground initially knows each task at release. The evaluator traverses communication windows in time order and transmits whole messages only: fragmentation is not implemented. A node can forward a task only after full reception; duplicate reception does not increase coverage. Candidate messages are ordered by priority, deadline, central-node preference, and target utility, then admitted only when residual window capacity and policy limits allow.

The nominal-original scenario uses the central-only/targeted policy, useful-recipient and deadline targeting, at most three forwarding hops, three observer copies, and five central-node copies. The *unlimited useful-deadline* diagnostic retains the useful-recipient and deadline definition but removes these hop and copy limits. It reveals latent architecture capability and must not be confused with the implementable nominal policy.

Accepted transfers retain task, sender, receiver, time, link type, message size, and hop depth. The algorithm is greedy and causal; it is not a globally optimized temporal multicommodity-flow solution.

4.7 Scenario Transformations

Fresh V2 evaluates 539 declared scenarios per case. Deterministic policy and task-size scenarios are complemented by repeated stochastic failures. A replicate is an independent rerun of the same architecture and severity with a different deterministic random seed; it measures variability due to which satellite, central node, link window, or outage interval is selected.

Random satellite and random central-node failures remove selected nodes, their incident links, and their observation opportunities. Targeted central-node failure removes nodes ranked by incident capacity and degree. Link-window outage removes selected window rows. Ground-station outage disables a sampled interval. Capacity degradation scales communication capacity. Combined stress varies task size and satellite failure together. Every transformed scenario reruns routing and observation evaluation; graph statistics alone are not accepted as robustness evidence.

The random seed is derived from the campaign seed, case identifier, scenario identifier, and replicate number using SHA-256. The first eight digest bytes define the numeric seed. Consequently, scenario outcomes are invariant to worker count and completion order.

4.8 Outcome and Resource Metrics

Observation fulfilment for N tasks is

$$F_{\text{obs}} = \frac{100}{N} \sum_{i=1}^N \mathbb{I}[t_i^{\text{obs}} \leq d_i]. \quad (4.6)$$

Let E_i be the useful-recipient set after the active failure transformation and $R_i(d_i)$ the unique useful recipients reached by deadline. Mean useful coverage uses

$$Q_i = 100 \frac{|R_i(d_i)|}{|E_i|}, \quad (4.7)$$

and strict dissemination is

$$S_{100,i}^{\text{useful}} = \mathbb{I}[E_i \neq \emptyset \wedge R_i(d_i) = E_i]. \quad (4.8)$$

The campaign stores original and surviving denominators separately so that failed nodes cannot create an artificial improvement. If complete dissemination occurs, $T_{100,i}$ is the release-to-last-useful-reception time. The reported P100 T_{100} is the worst completed task in the case; it is censored when any required completion is unavailable. It is not first-reception latency. The Fresh V2 canonical case table does not expose a separate first-reception field, so this report does not relabel T_{100} as first reception.

Communication burden is reported with transferred Mbit, transfer-event count, used-capacity percentage, central-node traffic share, peak node load, and a Gini coefficient over node-sent data. Performance and burden are reported as separate raw metrics before any composite score is introduced.

4.9 Checkpointing, Parquet Output, and Deterministic Resume

Each base architecture is one worker job that evaluates its eleven central-node cases. Scenario rows are written in batches of 20 to temporary files and atomically renamed when complete. On restart, the runner reads completed scenario identifiers and evaluates only the missing set. A `_SUCCESS.json` marker is created only after all expected scenario identifiers are present; exceptions create a `_FAILED.json` marker.

The tables are stored in Apache Parquet: a typed, columnar format with Zstandard compression. Parquet is intended for reliable analytical reading by Python, R, DuckDB, Spark, or similar tools rather than manual editing in a text editor. It preserves numeric types, compresses repeated columns efficiently, and allows column-selective reads. The principal case products are scenario metrics, task metrics, transfer events, central-node distribution, constellation nodes, manifests, and validation records.

Worker count is limited by the minimum of the configured maximum, operating-system CPU availability, and an available-memory estimate of approximately 3.00 GiB per worker. Changing 8 workers to 16 or 25 alters wall-clock time and the recorded execution fingerprint, but not the scientific seed or case definition. Scientific changes require a new output root; worker-only changes may safely resume the same root.

4.10 Validation before Inference

Post-processing follows a fixed gate sequence: read-only inventory, schema and traceability, completeness, independent metric recomputation, effect summaries, exact Pareto analysis, objective-weight sensitivity, robustness, and limitations. Duplicate scenario keys produced by interrupted resume are resolved deterministically before aggregation. Failed and censored observations remain visible rather than being silently dropped.

No ranking is performed until the relevant validation gates pass. The California Fresh V2 archive contains one incomplete case, which is excluded from comparisons that require all 539 scenarios. The integrity evidence is reported in Chapter 7 and Appendix C.

4.11 Pareto and Objective-Weight Method

The preferred architecture is treated as a decision conditional on mission priorities. First, exact dominance is evaluated on raw benefits and burdens. A case is Pareto efficient if no other case is at least as good on every objective and strictly better on one.

For the weight-consistent analysis, the performance pillar is

$$P = 0.25\tilde{F}_{\text{obs}} + 0.25\tilde{S}_{100} + 0.25\tilde{Q} + 0.25\tilde{T}, \quad (4.9)$$

where $\tilde{T}$ is inverse min-max P100 T_{100} and censored cases receive zero timeliness benefit rather than being removed. The resource-efficiency pillar is

$$R = 0.40(1 - \tilde{N}_{\text{sat}}) + 0.20(1 - \tilde{N}_{\text{CN}}) + 0.20(1 - \tilde{B}) + 0.20(1 - \tilde{E}), \quad (4.10)$$

where B is transferred data and E transfer-event count. The 40/20/20/20 split is a declared engineering preference: fleet size receives twice the weight of each remaining burden. It is not inferred from prices or claimed as universally correct.

The high-level utility is

$$U(\lambda) = \lambda P + (1 - \lambda)R, \quad 0 \leq \lambda \leq 1. \quad (4.11)$$

Sensitivity uses 10,000 reproducible draws with seed 20260825: $\lambda \sim \text{Uniform}(0, 1)$, performance weights from $\text{Dirichlet}(1, 1, 1, 1)$, and resource weights from $\text{Dirichlet}(4, 2, 2, 2)$. Winner frequency is a stability diagnostic over a declared preference distribution, not a probability that an architecture is objectively correct.

4.12 Implementation Ownership and Method Boundaries

Vincenzo Messina supplied the communication-model and TAT-C basis. The author implemented the wildfire-task processing, simulator integration, central-node placement, scenario runner, checkpoint/resume controls, validation, post-processing, statistical analysis, figures, and report.

The method evaluates task-message dissemination and follow-up opportunity under simplified orbit, sensing, and communications assumptions. It does not model image downlink, emergency alert delivery, operational response, monetary lifecycle cost, or a hardware-differentiated central-node subsystem. Conclusions therefore apply to the implemented communication/routing role and the tested task sets.

5 Experimental Design and Reproducibility

5.1 Two Evidence Layers

The thesis contains two validated but non-interchangeable result layers. Table 5.1 defines their role. The earlier nominal campaign is the only completed two-region evidence and supports the California–India contrast in Chapter 6. Fresh Full891 V2 adds scenario replay, robustness, task-size, policy, and objective-weight analysis, but only the California output is currently available. Numerical values are never averaged across these versions.

Table 5.1 Result layers and permitted use.

Layer	Coverage	Role in this report
Earlier nominal	891 California + 891 India architecture cases	Validated regional baseline and paired architecture comparison
Fresh Full891 V2	891 California case identifiers; 890 complete	Main scenario, robustness, task-size, policy, and decision evidence
Fresh Full891 V2 India	Run/validation pending	Required before Fresh V2 regional conclusions can be finalized

5.2 Common Architecture Grid

Both layers use the factorial grid in Table 5.2. Eleven central-node levels share each of 81 base architectures, allowing a within-geometry fraction sweep. The design contains 891 architecture identifiers per region.

Table 5.2 Architecture factors in the Full891 design.

Factor	Levels	Experimental role
Total satellites	60, 120, 180	Fleet scale and resource proxy
Orbital planes	3, 5, 10	Spatial distribution and contact geometry
Altitude	500, 800, 1000 km	Access/contact geometry
Inclination	60, 80, 98.6 deg	Regional access geometry
Central-node fraction	0, 10, ..., 100%	Degree of centralization
Walker phasing	$F = 1$	Fixed control

5.3 Earlier Two-Region Nominal Campaign

The earlier campaign contains one deterministic nominal evaluation for every architecture in California and India, for 1,782 analytical case rows. All expected cases passed completeness and configuration checks. Ten case metrics were independently recomputed, giving 17,820 comparisons and no failure

at tolerance 10^{-6} . This layer predates the Fresh V2 scenario contract and is retained because it supplies the completed regional comparison requested in the research questions.

Its interpretation is deliberately limited. It supports statements about the observed two-event baseline under its frozen configuration, not Fresh V2 failure retention or task-size sensitivity. Chapter 6 labels every table and figure from this layer.

5.4 Fresh Full891 V2 Scenario Matrix

Fresh V2 creates 539 scenario definitions for each architecture. Table 5.3 shows the construction. Stochastic families use 30 replicates per severity. Deterministic families are evaluated once per level.

Table 5.3 Fresh V2 scenario contract for each architecture case.

Scenario family	Construction	Rows
Core, policy, and unlimited	Seven named scenarios	7
Task size	$0.5\times, 1\times, 2\times, 4\times$	4
Random satellite failure	Four levels $\times$ 30 replicates	120
Random central-node failure	Four levels $\times$ 30 replicates	120
Link-window outage	Four levels $\times$ 30 replicates	120
Ground-station outage	Four levels $\times$ 30 replicates	120
Targeted central-node failure	Four deterministic levels	4
Capacity degradation	Four deterministic levels	4
Combined stress	Two sizes $\times$ two satellite-failure levels $\times$ 10 replicates	40
Total		539

Here a replicate means a repeated random trial at the same architecture, scenario family, and severity. The architecture and tasks remain fixed, but the failed nodes, removed links, or outage interval can differ. Thirty replicates estimate the distribution of a random-failure response; they are not 30 different architectures.

If both regions complete the contract, the design contains $1,782 \times 539 = 960,498$ expected scenario rows. CN0 has no central nodes, so 124 central-node-failure rows per CN0 case are explicitly NOT_APPLICABLE; the two-region expected total is 20,088. These rows remain in the manifest and are not treated as failures or silently dropped.

5.5 California Fresh V2 Integrity Audit

The California archive was inspected before numerical aggregation. Table 5.4 records the result. The expected full total is $891 \times 539 = 480,249$ rows. Interrupted/resumed batches produced 4,269 duplicate scenario keys; deterministic de-duplication retains the final unique record for each declared key.

The incomplete identifier is

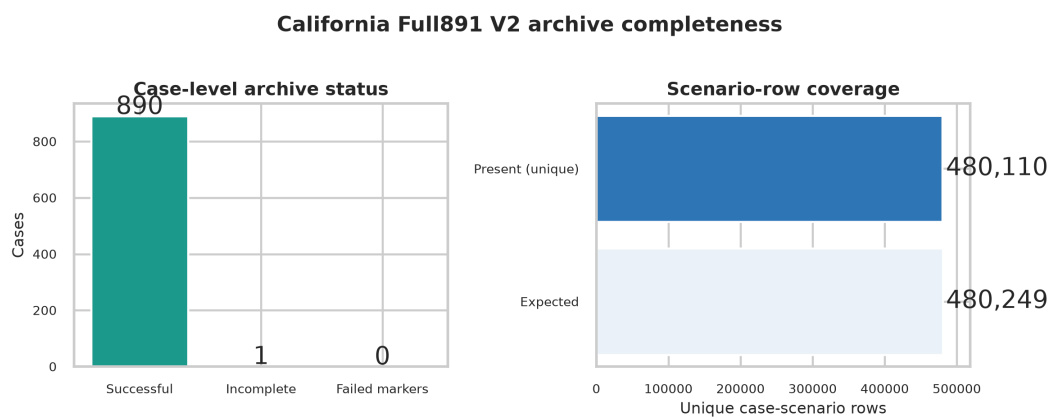
T180_P10_H1000_I0986_CN100_F1.

It contains 400 of 539 rows and lacks 99 ground-station-outage and 40 combined-stress evaluations. It is retained in the inventory but excluded whenever a complete scenario comparison is required. This leaves 890 complete cases. Eighty of the 81 base geometries have a fully balanced 11-level CN sweep, giving 880 cases for balanced central-node marginal analysis.

Two configuration fingerprints occur: one for 889 cases and one for two cases. Field-level comparison attributes the difference to worker/runtime provenance, not tasks, scientific configuration, seeds, or scenario definitions. The worker change is therefore reported but does not invalidate the science.

Table 5.4 California Fresh V2 campaign integrity.

Quantity	Value
Base-architecture directories	81
Architecture case directories	891
Complete success markers	890
Failure markers	0
Expected scenarios per complete case	539
Raw scenario rows	484,379
Unique rows across all cases	480,110
Resume duplicates removed	4,269
Unique rows in success cases	479,710
Completed / not-applicable / failed status rows	469,666 / 10,044 / 0
Architecture-validation checks passed	19,179 of 19,179

**Figure 5.1** California Fresh V2 completeness and scenario-family coverage after deterministic de-duplication.

5.6 Validation Gates and Claim Freeze

The checking method applies the following gates in order:

1. inventory paths, manifests, markers, and expected identifiers without altering raw output;
2. confirm schema, types, join keys, task provenance, and scenario traceability;
3. check completeness, uniqueness, status, and explicit not-applicable rows;
4. recompute metrics from lower-level task/transfer evidence where retained;
5. calculate paired effects and architecture sweeps only on eligible cases;
6. construct the exact Pareto set before applying weights;
7. run objective-weight and robustness sensitivity with declared seeds; and
8. write a claim ledger containing both supported conclusions and unavailable evidence.

This order prevents a visually plausible ranking from masking missing cases, duplicate resume rows, a changed denominator, or censored latency. Raw observations remain visible beside normalized scores. A result is called robust only after the physical/temporal scenario has been replayed; a nominal task failure label is not a robustness experiment.

5.7 Analysis Populations

Different questions require different subsets:

- all 891 identifiers for archive inventory;
- 890 complete cases for main California Fresh V2 summaries;
- 880 cases in 80 complete base sweeps for balanced CN-fraction curves;
- eligible cases within each scenario family after explicit NOT_APPLICABLE handling;
- 48 cases on the original six-objective unlimited Pareto set; and
- 108 cases on the recomputed eight-objective frontier used for weight analysis.

The population is stated with each analysis. This is particularly important for conditional P100 T_{100} , which exists for 824 of the 890 unlimited cases, and for CN-failure scenarios, which do not apply to CN0.

5.8 Output Contract and Post-Processing Inputs

Each case folder contains a manifest, architecture validation, central-node distribution, constellation nodes, compressed scenario metrics, task metrics, and selected transfer events. Scenario and task tables are partitioned into files such as `case_metrics_part_0000.parquet`. A success marker closes the case only after the expected scenario identifiers are present.

The post-processing release included with this report contains compact CSV/JSON summaries and figures rather than the full raw Parquet archive. The compact files preserve values used in the thesis and can be inspected with ordinary spreadsheet or text tools. The raw archive remains external because it is too large for normal Git history. Its directory and marker structure is described in Appendix E.

5.9 Statistical and Decision Analysis

Central-node curves aggregate within complete base sweeps. Non-CN factor summaries are descriptive marginal means, and effect sizes are unadjusted one-factor η^2 values; neither is interpreted as a causal coefficient in the presence of interactions. Spearman correlation describes monotonic association only.

Robustness is reported primarily as paired percentage-point change from the same architecture's unlimited baseline. Replicate distributions are summarized across the 30 seeded draws. Task-size sensitivity compares the four deterministic multipliers. Policy ablation compares central, peer, and hybrid implementations, with implementation-equivalent outputs called out rather than counted as independent evidence.

The decision sequence is: apply mission feasibility thresholds, compute exact Pareto efficiency, show the raw performance/resource vector, apply declared weights, and vary those weights. Pure resource weighting may select a zero-performance case; pure performance weighting may select an extreme high-resource case. These endpoints demonstrate why weighted utility cannot substitute for engineering constraints.

5.10 Compute Configuration and Resume Semantics

The VM runner limits concurrent workers using CPU availability and free memory. A planning allowance of approximately 3.00 GiB per worker prevents a nominal core count from exhausting RAM. A 16-worker setting is therefore safe only when the VM retains sufficient memory for the operating system and worker peaks.

Stopping between completed checkpoints does not invalidate completed scenario rows. On restart, the program recognizes completed identifiers and continues the missing set. Abrupt interruption can leave temporary or duplicate batch records, which is why post-processing de-duplicates declared keys and requires the success marker. Editing worker count does not accelerate workers already running; the new value takes effect only for a restarted/resumed launcher.

5.11 Evidence Still Missing

The Fresh V2 India output was still running or unavailable at the report freeze. The fixed-radius California archive also contains no observation-radius sensitivity, and its canonical case table contains no separate first-reception-latency field. These absences are recorded as limits, not filled from the earlier nominal layer. Once India passes the same gates, the regional comparison should be repeated on paired architecture identifiers and normalized per task.

6 Validated Two-Region Nominal Baseline

6.1 Evidence-Layer Boundary

This chapter reports the earlier validated nominal campaign: 891 California and 891 India architecture cases. It is retained because it is the completed two-region evidence layer. Its values are not Fresh Full891 V2 values and are not numerically combined with Chapter 7. The purpose of this chapter is to establish the regional baseline and show which architecture rankings transfer between the two event sets under the earlier frozen model.

6.2 Result Status

All results in this chapter passed the validation gates defined in Chapter 5. They supersede the numerical conclusions of still earlier V10, V16, V19, 30-hour screening, and fixed-architecture slide-deck experiments. They do not supersede Fresh V2; instead, Chapter 7 reports that later California scenario layer separately.

The analytical population contains 891 cases for California and 891 for India. Every case has the expected task count, and every required case status is completed. Core summary metrics were recomputed from lower-level outputs in 17,820 comparisons with zero failures at a tolerance of 10^{-6} . The numerical evidence is therefore internally consistent even though external model-validity limitations remain.

6.3 Regional Campaign Summary

Table 6.1 summarizes the full architecture population. India has a higher mean observation fulfilment but a lower mean useful-receiver fraction. Transfer events per task are nearly identical, whereas transferred data per task is higher for California because its priority mix contains proportionally more large urgent packets.

Table 6.1 Region-normalized summary across all 891 cases per event region.

Region	Obs. (%)	S_{100}^{useful} (%)	Receivers (%)	Mbit/task	Events/task
California	47.468	0.624	13.868	5.884	6.186
India	55.066	0.462	12.060	4.620	6.202

The very low mean S_{100}^{useful} values require careful interpretation. They do not mean that almost no task is observed: mean observation fulfilment is roughly 47–55 percent. They mean that completing dissemination to every simulator-defined useful recipient before each deadline is rare under the targeted policy, three-hop limit, and copy limits. Observation requires only one suitably timed observer, while S_{100}^{useful} requires all useful recipients.

6.4 Central-Node Fraction Effects

Figure 6.1 contains the four principal central-node marginal responses. Each point averages 81 base architectures. The top-left panel shows a large gain when moving from 0 to 10 percent central nodes, followed by a slower increase and a decrease at 100 percent. The bottom-right panel shows an almost monotonic and strongly nonlinear latency reduction. The dissemination and receiver panels reveal the opposing effect: broad reach is highest at modest central-node fractions and decreases at high fractions.

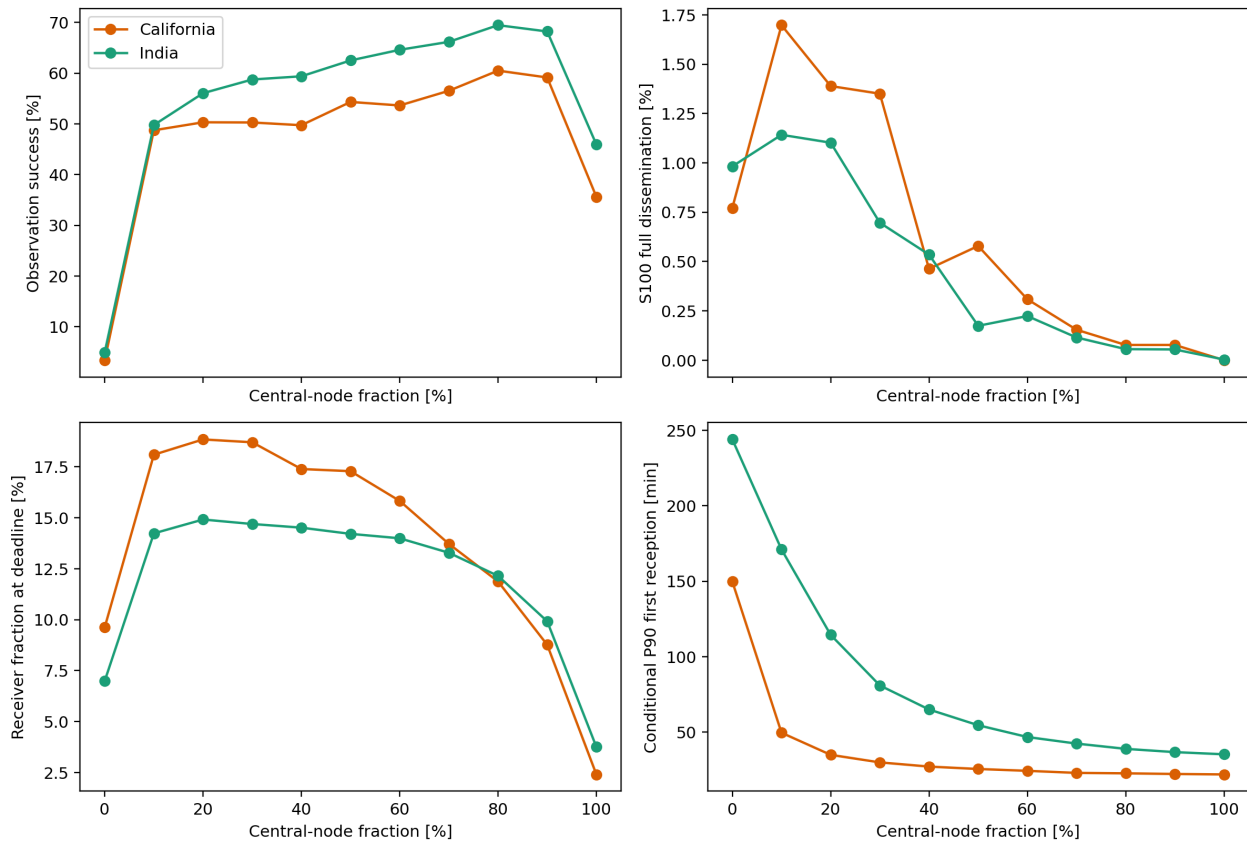


Figure 6.1 Regional marginal response to central-node fraction across the full final grid. S_{100}^{useful} denotes complete dissemination to all simulator-defined useful recipients.

Selected levels are reported in Table 6.2. The table illustrates the key engineering trade. For California, moving from CN0 to CN10 raises mean observation fulfilment from 3.40 to 48.73 percent and reduces P90 first-reception latency from 149.78 to 49.65 minutes. For India, the corresponding observation change is 4.92 to 49.77 percent and the latency change is 243.94 to 171.06 minutes. Further centralization continues to improve conditional first reception, but useful-recipient breadth does not continue to improve.

6.4.1 Observation fulfilment

The region-aggregated observation maximum occurs at 80 percent central nodes: 60.49 percent in California and 69.46 percent in India. At 90 percent, both regional means are slightly lower, and at 100 percent they fall to 35.61 and 45.92 percent. The result does not support a monotonic “more central nodes is better” claim.

The highest individual California observation result is 93.75 percent for T060_P05_H0800_I0800_CN050_F1. The highest India result is 99.094 percent for T180_P10_H1000_I0600_CN090_F1. Nei-

Table 6.2 Selected regional central-node marginal means across 81 base architectures per row.

Region	CN (%)	Obs. (%)	S_{100}^{useful} (%)	Receivers (%)	P90 first (min)
California	0	3.395	0.772	9.623	149.779
California	10	48.727	1.698	18.098	49.648
California	20	50.309	1.389	18.847	35.013
California	50	54.321	0.579	17.289	25.647
California	80	60.494	0.077	11.874	22.768
California	100	35.610	0.000	2.401	22.039
India	0	4.921	0.982	6.988	243.944
India	10	49.766	1.142	14.235	171.061
India	20	56.044	1.102	14.916	114.486
India	50	62.503	0.174	14.211	54.628
India	80	69.460	0.056	12.148	38.982
India	100	45.925	0.003	3.752	35.365

ther of these cases completes S_{100}^{useful} for any task, emphasizing that a case can observe nearly all tasks without disseminating them to every useful recipient.

6.4.2 Useful-recipient completion and receiver coverage

Mean S_{100}^{useful} is small at every aggregate level and reaches its marginal maximum at CN10 in both regions: 1.698 percent for California and 1.142 percent for India. Mean useful-receiver coverage instead peaks at CN20: 18.85 percent for California and 14.92 percent for India. Beyond this point, the number of useful receivers reached by the deadline decreases, particularly at CN100.

The strongest individual S_{100}^{useful} cases are lower-resource 60-satellite configurations. California reaches 25.0 percent for T060_P03_H0800_I0986_CN030_F1; India reaches 21.475 percent for T060_P05_H1000_I0600_CN020_F1. These same cases also have the largest mean receiver fraction in their region, 47.58 and 45.56 percent, respectively.

The fact that 60-satellite cases dominate this metric is partly mathematical and partly operational. With fewer eligible recipients, completion requires fewer unique arrivals. Larger constellations create more possible observers but also enlarge the dissemination target. This is why S_{100}^{useful} should be compared together with observation fulfilment and resource burden rather than used alone.

6.4.3 First-reception latency

Conditional P90 first-reception latency falls from 149.78 to 22.04 minutes in California and from 243.94 to 35.36 minutes in India between CN0 and CN100. Most of the California improvement occurs by CN20, while the India response continues to fall strongly through CN50. The later reduction has diminishing magnitude in both regions.

The smallest per-case conditional P90 is 1.40 minutes in California and 8.52 minutes in India. These values are not labelled “best latency architectures” because conditional latency can look favourable when many difficult tasks are censored. The completion fraction must accompany any latency rank. This is the practical reason for retaining censored-task columns in the canonical tables.

6.5 Architecture Factor Effects

Table 6.3 collects the most decision-relevant factor-level means. Each entry pools the other factors, so it describes association within the grid rather than a controlled one-factor experiment. The table shows why geometry and centrality cannot be reduced to a single monotonic rule.

Table 6.3 Selected factor-level marginal effects across the full nominal grid.

Region	Factor	Level and interpretation	Mean obs. (%)	Mean P90 first (min)
California	Satellites	60; best mean observation, widest receiver denominator advantage	51.04	44.63
California	Planes	5; highest observation mean	55.16	27.55
California	Inclination	60 deg; highest observation and lowest latency	54.02	13.23
California	CN fraction	80%; highest mean observation	60.49	22.77
India	Satellites	120; marginally highest observation mean	55.57	78.79
India	Planes	10; highest observation mean	62.27	68.98
India	Inclination	60 deg; strongest regional geometry	72.42	54.22
India	CN fraction	80%; highest mean observation	69.46	38.98

Increasing satellite count reduces the conditional P90 of first reception but decreases useful-receiver fraction. In California, mean receiver coverage falls from 20.95 percent at 60 satellites to 8.55 percent at 180. In India it falls from 18.33 to 7.32 percent. Again, this is not evidence that larger constellations communicate less; it shows that the target recipient set grows faster than the bounded-copy forwarding policy.

Plane count affects regions differently. California has its highest mean observation at five planes, while India benefits strongly from ten. Inclination produces the largest regional contrast: 60 degrees yields 72.42 percent mean observation in India, whereas 98.6 degrees yields 40.43 percent. This is consistent with the task-region latitude and the event epoch, but the simulator does not isolate which geometrical component is responsible.

Altitude has weaker mean association with observation fulfilment than inclination or plane count. Higher altitude reduces conditional P90 first reception in both regions, but the observation means differ by only a few percentage points across 500, 800, and 1000 km after pooling other factors.

6.6 Exploratory Correlation

The strongest Spearman associations are reported in Table 6.4. Central-node fraction has a negative association with P90 first-reception latency in both regions, consistent with faster first delivery. Satellite count has the strongest negative association with receiver fraction. India observation fulfilment is strongly negatively associated with inclination in the sampled levels.

Table 6.4 Selected exploratory Spearman associations across 891 cases per region.

Region	Factor	Metric	ρ
California	Central-node fraction	P90 first-reception latency	−0.553
California	Satellite count	Mean useful-receiver fraction	−0.555
California	Plane count	Observation fulfilment	+0.390
California	Central-node fraction	Observation fulfilment	+0.285
India	Central-node fraction	P90 first-reception latency	−0.686
India	Satellite count	Mean useful-receiver fraction	−0.614
India	Inclination	Observation fulfilment	−0.586
India	Central-node fraction	Observation fulfilment	+0.336

No causal statement is attached to these coefficients. Central-node fraction and satellite count change the denominator and routing opportunities, and all coefficients pool the other factors. The full-factorial structure supports controlled marginal contrasts, but a mechanistic causal model would require interaction terms and possibly additional replicates.

6.7 Exploratory Regression of Central-Node Marginal Means

Figure 6.2 compares the best leave-one-level-out model for each regional response. An exponential-with-offset model describes the steep latency fall better than the polynomial candidates, with leave-one-level-out RMSE of 23.57 minutes for California and 6.29 minutes for India. Quadratic models are preferred for receiver fraction in both regions, capturing the intermediate peak and later decrease.

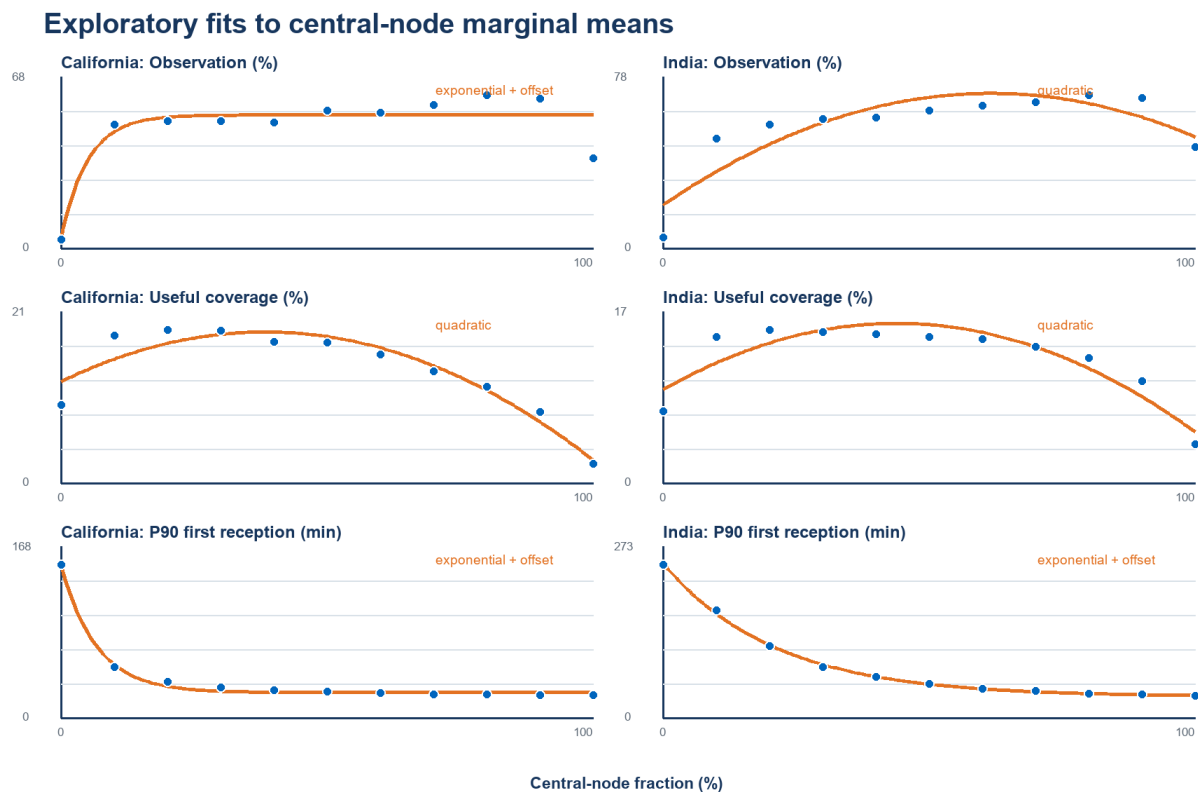


Figure 6.2 Exploratory regression of the 11 regional central-node marginal means. The displayed model minimizes leave-one-level-out RMSE among linear, quadratic, cubic, and exponential-with-offset candidates.

Table 6.5 gives the selected models. California observation has a large cross-validated error despite a high in-sample R^2 , because a saturating exponential does not reproduce the CN100 decline. This is a warning against treating the fitted curve as a physical law. India observation and both receiver responses are better represented by a quadratic over the sampled range, but extrapolation beyond 0–100 percent is meaningless.

Table 6.5 Best exploratory regression model by leave-one-central-node-level-out error.

Region	Response	Selected model	R^2	LOOCV RMSE
California	Observation fulfilment	Exponential + offset	0.833	14.514
California	Useful-receiver fraction	Quadratic	0.911	2.611
California	P90 first reception	Exponential + offset	0.995	23.567 min
India	Observation fulfilment	Quadratic	0.778	14.210
India	Useful-receiver fraction	Quadratic	0.851	2.310
India	P90 first reception	Exponential + offset	0.999	6.294 min

The regression result supports two qualitative conclusions. Latency benefit saturates, while receiver breadth is hump-shaped. It does not identify an optimum by itself because observation, receiver coverage, latency, and resource burden peak at different points.

6.8 Paired Regional Comparison

Figure 6.3 compares observation fulfilment for identical architecture identifiers under the two event regions. Points above the diagonal favour India and points below favour California. The broad scatter shows that an architecture’s regional performance cannot be inferred from one event alone.

Rank agreement is reported in Table 6.6. Conditional P90 first reception and receiver fraction are comparatively stable, whereas S_{100}^{useful} ranking agreement is only 0.110. This low agreement is consistent with the different task counts, priority mixes, deadlines, and useful-recipient completion rates.

Table 6.6 Spearman rank agreement between California and India over 891 paired architecture identifiers.

Metric	Rank agreement
Observation fulfilment	0.504
S_{100}^{useful}	0.110
Mean useful-receiver fraction	0.738
Conditional P90 first-reception latency	0.775

6.9 Exact Pareto Sets

Figure 6.4 plots observation fulfilment against the declared resource-burden proxy and outlines exact Pareto members computed in the full five-objective space. A point can be Pareto optimal even if it appears dominated in this two-dimensional projection, because it may provide better S_{100}^{useful} , receiver coverage, or conditional latency.

California has 183 Pareto cases and India has 142, corresponding to 20.5 and 15.9 percent of their candidate sets. The relatively large fronts show that the objectives are genuinely conflicting. Collapsing them into a single equal-weight score would discard viable design choices and could make the recommendation sensitive to arbitrary scaling.

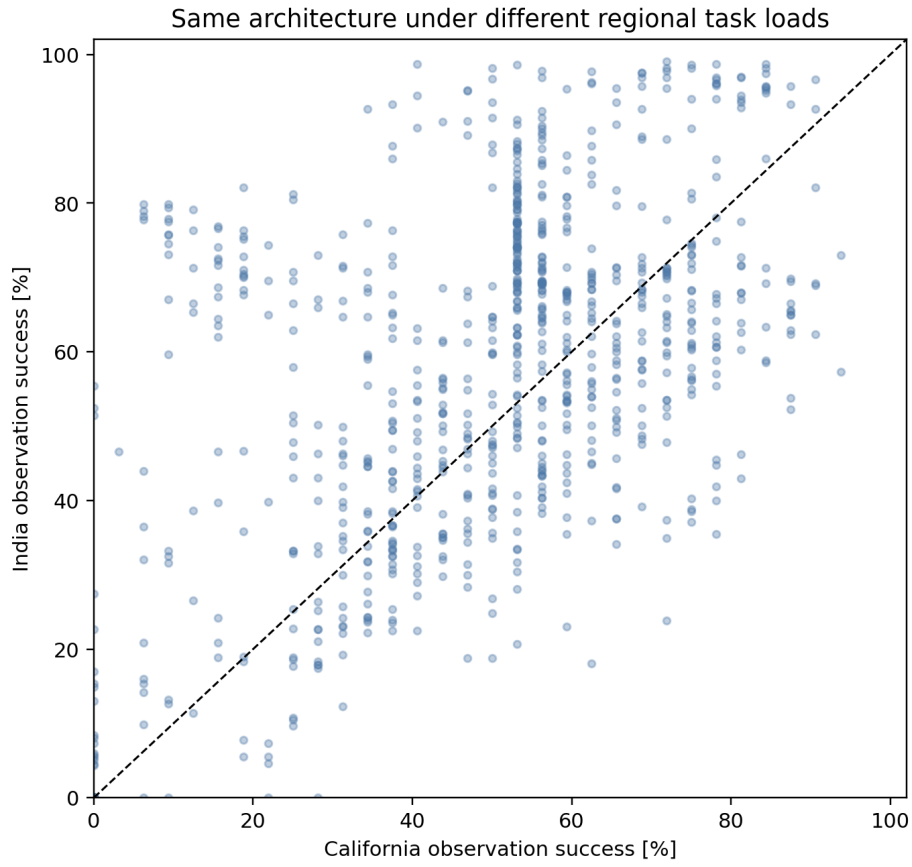


Figure 6.3 Paired observation fulfilment for identical architecture identifiers under the California and India task sets.

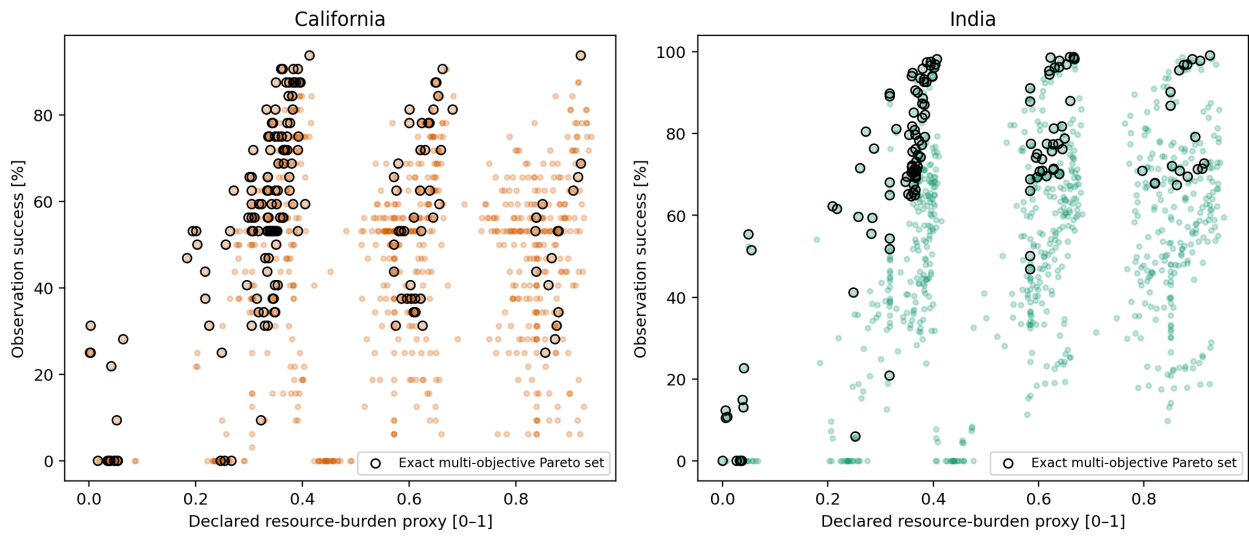


Figure 6.4 Observation fulfilment versus non-monetary resource burden. Black outlines identify cases on the exact five-objective Pareto set.

6.10 Nominal Result Synthesis

The nominal campaign supports four principal findings:

1. introducing a modest central-node fraction produces a large improvement over CN0 in observation fulfilment and first-reception latency under the implemented central-only topology;
2. further centralization continues to reduce first-reception latency but does not monotonically improve observation, useful-recipient coverage, or S_{100}^{useful} ;
3. regional workload and geometry change the architecture ranking, so no winner is independent of the region; and
4. the bounded-copy, three-hop targeted policy makes rapid delivery and complete useful-recipient dissemination distinct objectives.

These findings answer the earlier nominal centrality question but do not establish failure tolerance. Chapter 7 therefore treats robustness as a separate Fresh V2 replay and reports the validated California values independently.

7 Full891 Fresh V2 California Results and Sensitivity

7.1 Scope of the Fresh V2 Results

This chapter is the principal Fresh Full891 V2 result layer. It uses the 890 complete California architectures and, for central-node marginal curves, the 880 cases belonging to 80 complete eleven-level sweeps. The omitted case and all scenario counts are documented in Chapter 5 and Appendix C. India Fresh V2 is not yet available; no California value in this chapter is presented as an India result.

Two scenarios anchor the interpretation. *Nominal original* is the bounded central-only targeted policy with the operational deadline, three-hop limit, three observer copies, and five central-node copies. *Unlimited useful deadline* removes the hop/copy limits while retaining the useful-recipient and deadline definitions. The second scenario is a diagnostic upper-capability replay, not a directly recommended operating mode.

7.2 Bounded Policy versus Architecture Potential

Table 7.1 quantifies the gap between the two anchor scenarios. Nominal mean observation fulfilment is already 61.59 percent, but strict useful-recipient completion is only 0.63 percent and mean useful coverage only 15.29 percent. Removing copy and hop limits raises strict completion to 74.69 percent and coverage to 91.14 percent. The increase requires more than ten times the transferred data.

Table 7.1 California mean outcome over 890 complete Fresh V2 architectures.

Scenario	Obs. (%)	S_{100}^{useful} (%)	Coverage (%)	Data (Mbit/case)
Nominal original	61.59	0.63	15.29	197.55
Unlimited useful deadline	77.62	74.69	91.14	2,047.57
Difference	+16.04	+74.07	+75.85	+1,850.02

The main explanation is operational rather than purely geometric. The bounded policy frequently reaches a useful observer, which can complete the observation objective, but does not distribute the task to every useful recipient. The diagnostic replay uses additional copies and deeper temporal paths. Therefore, the low nominal S_{100}^{useful} is a property of the architecture–policy combination, not evidence that the constellation has no communication opportunities.

7.3 Central-Node Fraction

Figure 7.1 presents both operating regimes. Under the nominal policy, mean observation fulfilment increases sharply from 29.14 percent at CN0 to 64.88 percent at CN10, reaches 71.13 percent at CN50, and falls to 35.70 percent at CN100. Strict dissemination peaks at 1.72 percent at CN10, while mean useful coverage peaks around CN20 at approximately 20.7 percent and falls to 2.42 percent at CN100. Intermediate centralization is therefore preferable under the bounded policy.

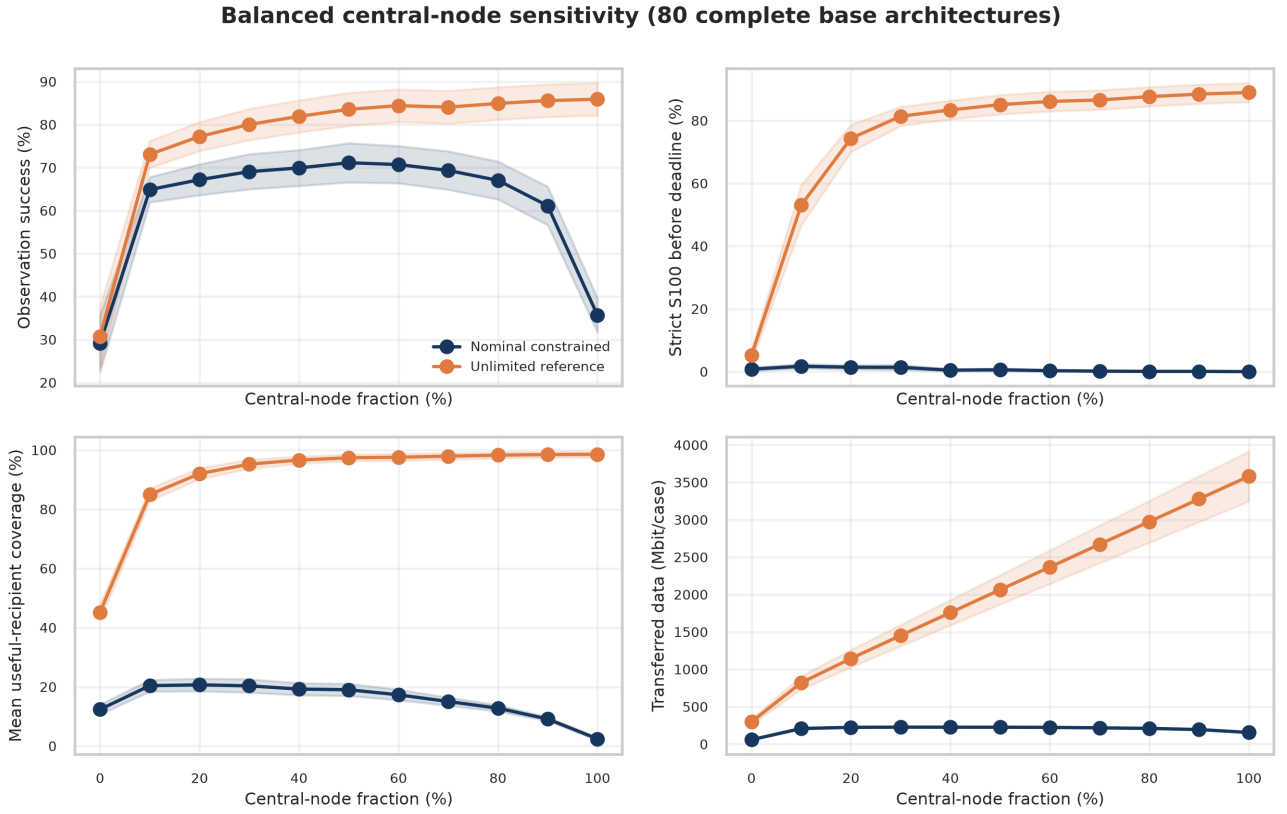


Figure 7.1 Central-node fraction response across 80 balanced California Fresh V2 base architectures. Shaded or interval displays summarize variation across base geometries.

The unlimited replay reveals a different, saturating regime. Selected marginal means are shown in Table 7.2. S_{100}^{useful} grows rapidly up to the intermediate range and then provides smaller incremental gains, while traffic continues to grow. From CN20 to CN100, mean S_{100}^{useful} increases by about 14.6 points, but transferred data increases by approximately 2,429 Mbit per case (about 211 percent).

Table 7.2 Selected unlimited useful-deadline central-node marginal means.

CN (%)	S_{100}^{useful} (%)	Coverage (%)	P100 T_{100} (min)	Data (Mbit)	Events
0	5.16	45.21	232.6	297	407
20	74.34	92.02	120.0	1,152	1,333
40	83.40	96.61	97.4	1,771	1,939
70	86.56	—	—	—	—
100	88.95	98.55	73.2	3,581	3,709

The dashes denote values not selected for the compact table; they are not treated as zero. P100 T_{100} is conditional and especially censored at CN0. In the nominal scenario, mean maximum routing depth reaches the configured limit of three by approximately CN20. In the unlimited replay it rises from 1.0 at CN0 to about 7.8 at CN20, 9.6 at CN40, and 11.8 at CN100. Part of the unlimited gain is therefore attributable to policy freedom, not central-node fraction alone.

7.4 Architecture Factors and Dependence

Figure 7.2 shows descriptive main effects. Within the unlimited balanced subset, unadjusted one-factor η^2 for CN fraction is 0.690 for S_{100}^{useful} and 0.820 for useful coverage. Plane count is the next

largest reported factor for S_{100}^{useful} at 0.112. For observation fulfilment, CN fraction and plane count have η^2 values of 0.402 and 0.361. Satellite count, altitude, and inclination are smaller in these marginal summaries.

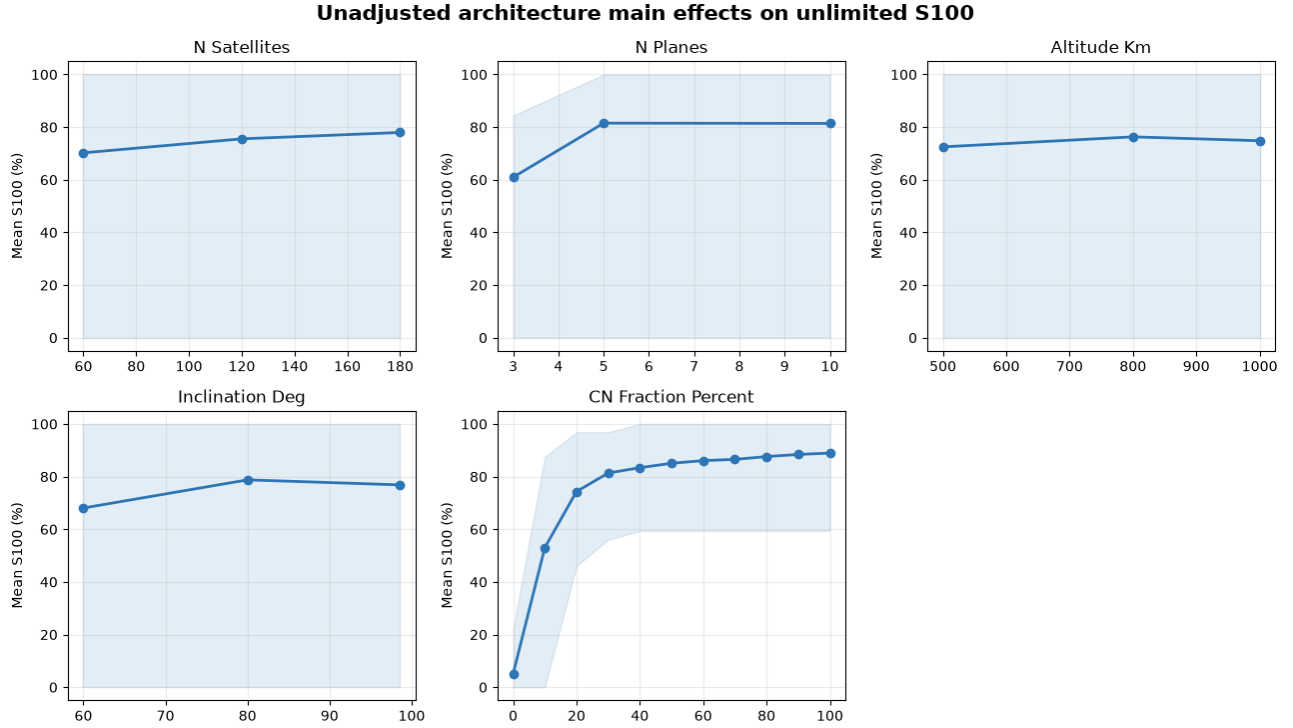


Figure 7.2 California Fresh V2 marginal architecture effects under the unlimited useful-deadline replay.

These values show that dissemination depends strongly on the central-node role, while observation also depends materially on plane geometry. Because the design contains interactions and discrete placement changes, the unadjusted effects are descriptive. They do not imply that changing one factor while holding an unmodelled system fixed will reproduce the marginal mean.

7.5 Policy Ablation

The policy ablation in Table 7.3 separates central-only access from all-peer link availability. Peer-only and hybrid produce identical recorded outcomes in the current implementation. This identity is important: it indicates that the implemented hybrid preference does not change admitted transfers relative to peer-only for this scenario set. It is not evidence that every possible hybrid algorithm is equivalent to peer routing.

Table 7.3 Mean California policy-ablation outcome over 890 complete architectures.

Policy scenario	Obs. (%)	S_{100}^{useful} (%)	Coverage (%)	Data (Mbit/case)
Central / unlimited	77.62	74.69	91.14	2,047.57
Peer	83.04	85.44	97.20	3,485.72
Hybrid	83.04	85.44	97.20	3,485.72

The peer-capable variants improve observation by 5.42 percentage points and S_{100}^{useful} by 10.75 points relative to central-only unlimited routing, but increase mean transferred data by approximately 1,438 Mbit per case. The comparison demonstrates that link availability and central-node fraction must be interpreted jointly.

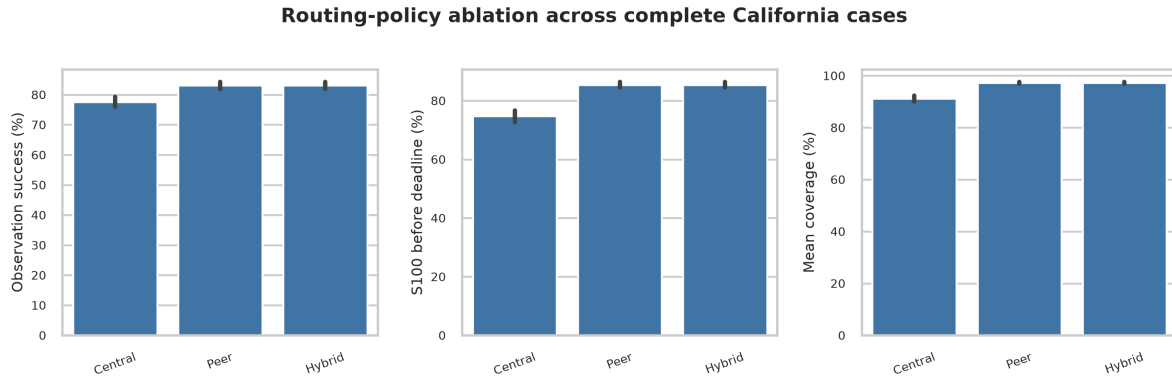


Figure 7.3 Policy ablation: performance gain and communication burden under central, peer, and hybrid link use.

7.6 Task-Size Sensitivity

Task-size scaling provides a direct communication-load stress test while retaining the same tasks and physical opportunities. Table 7.4 and Figure 7.4 show monotonic degradation as packets grow. Halving the packet size slightly improves performance and approximately halves traffic; quadrupling it reduces observation fulfilment by 8.14 points and strict completion by 14.59 points relative to the $1\times$ case.

Table 7.4 California task-size sensitivity over 890 complete architectures.

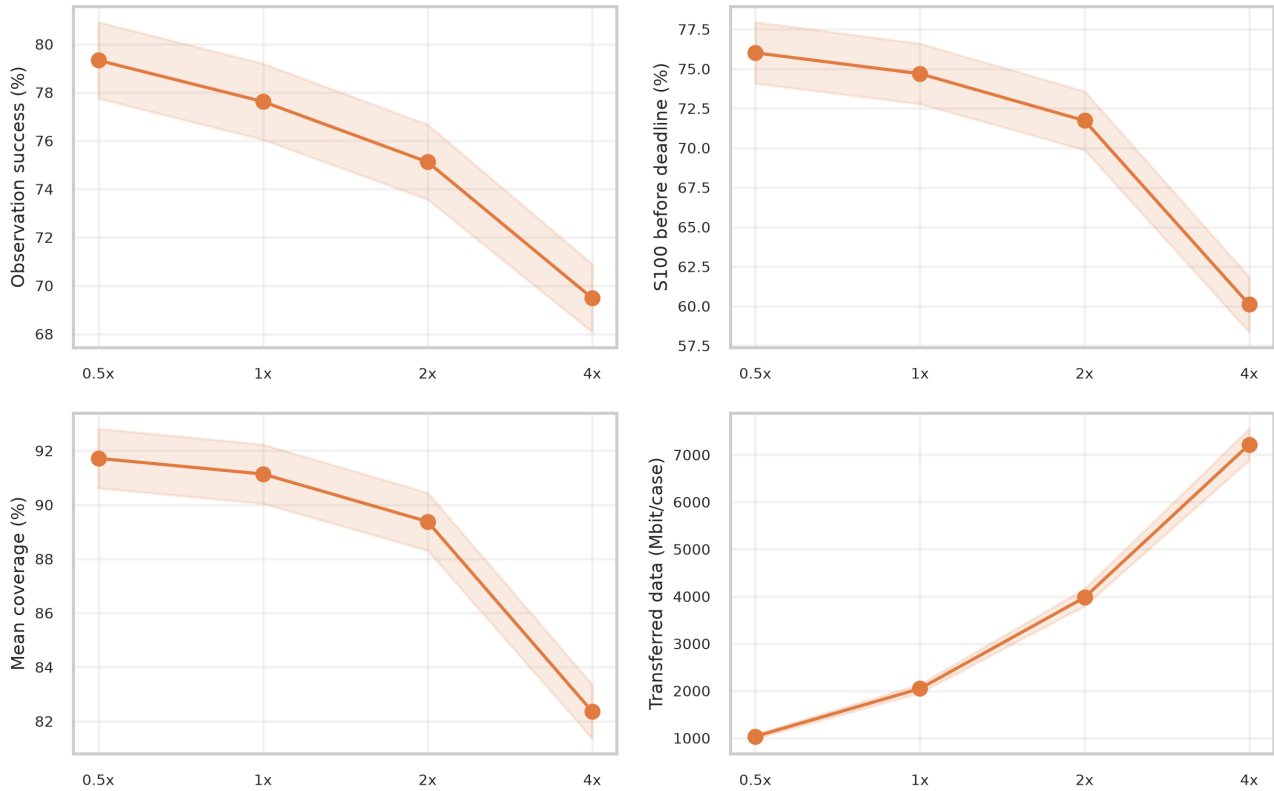
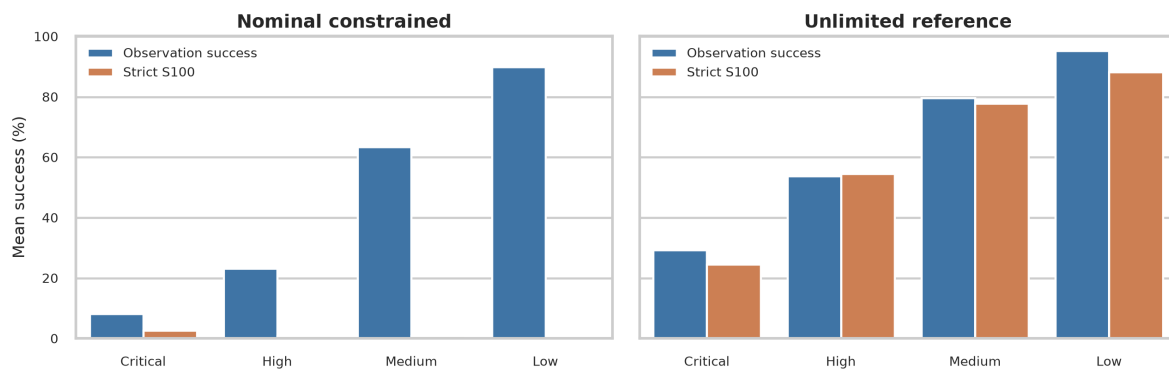
Multiplier	Obs. (%)	S_{100}^{useful} (%)	Coverage (%)	Data (Mbit/case)
0.5×	79.33	76.01	91.72	1,036.55
1×	77.62	74.69	91.14	2,047.57
2×	75.12	71.73	89.38	3,979.20
4×	69.48	60.10	82.36	7,219.41

The balanced 40-percent-CN candidate retains 96.875 percent observation at $2\times$ size but falls to 87.5 percent at $4\times$; its S_{100}^{useful} changes from 100 percent at $1\times$ to 96.875 and 78.125 percent. A recommendation based on small task descriptions is therefore not invariant to heavier products.

7.7 Priority-Class Outcomes

California contains one Critical, six High, sixteen Medium, and nine Low tasks. Figure 7.5 shows that urgent classes are substantially harder under both anchor scenarios. The nominal mean observation outcomes are 8.20, 23.09, 63.43, and 89.90 percent for Critical through Low. Under the unlimited replay they become 29.21, 53.75, 79.66, and 95.31 percent. Corresponding unlimited S_{100}^{useful} means are 24.49, 54.55, 77.80, and 88.16 percent.

This pattern does not isolate a causal priority effect. Priority is jointly encoded through deadline and message size, and each class contains different task locations and release times. The Critical class has one task. The result supports a workload-heterogeneity conclusion, not a general claim that high-FRP fires are intrinsically harder to observe.

Task-size sensitivity under unlimited routing**Figure 7.4** Task-size sensitivity of performance and communication burden.**Priority-class outcomes (1 critical, 6 high, 16 medium, 9 low tasks per case)****Figure 7.5** California task-priority outcomes under nominal and unlimited scenarios.

7.8 Failure Robustness

For failure scenarios, each architecture is compared with its own unlimited baseline. Table 7.5 reports the average percentage-point effect at the 30-percent level. Ground-station outage is the dominant tested stressor. Satellite and central-node failure produce moderate average losses; capacity degradation is smaller at the tested level.

Table 7.5 Mean paired performance change at the 30-percent stress level.

Failure family	Δ Obs. (pp)	$\Delta S_{100}^{\text{useful}}$ (pp)	Interpretation
Ground-station outage	-21.46	-20.84	Dominant tested vulnerability
Targeted central-node failure	-2.95	-6.57	Larger dissemination than observation penalty
Random central-node failure	-2.86	-5.98	Moderate mean loss
Random satellite failure	-3.01	-3.91	Geometry and forwarding loss
Capacity degradation	< -1	approximately -1	Small at tested severity

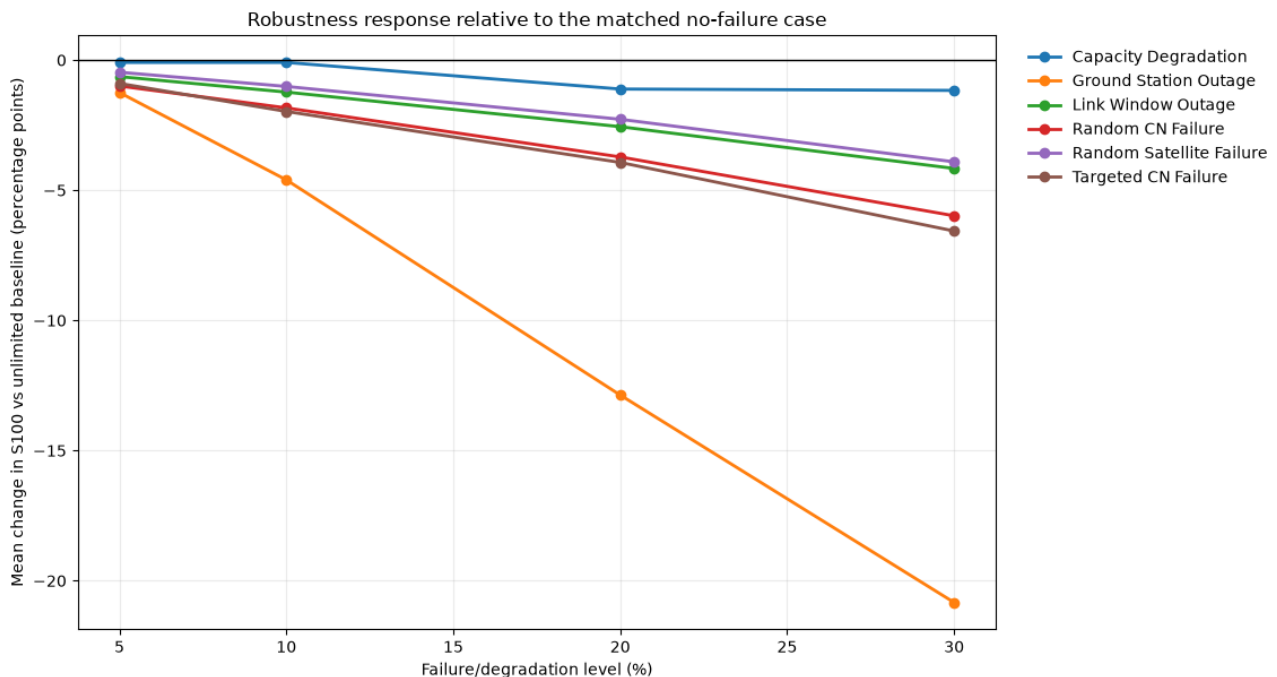


Figure 7.6 Paired change in strict useful dissemination across Fresh V2 robustness families and severities.

For the balanced candidate T060_P10_H0800_I0986_CN040_F1, the unlimited baseline is 96.875 percent observation and 100 percent S_{100}^{useful} . At 30-percent random central-node failure, replicate means are 91.25 and 93.75 percent; at 30-percent random satellite failure they are 92.08 and 95.52 percent. Deterministic targeted-CN failure retains 93.75 percent observation and 100 percent S_{100}^{useful} , although P100 T_{100} increases from 66.10 to 124.42 minutes. Ground-station-outage replicate means fall to 66.77 and 69.79 percent. Ground diversity is therefore a more direct design need than simply raising the central-node fraction beyond the intermediate range.

7.9 Pareto Trade Space

No architecture dominates all others on performance and burden. Figure 7.7 shows the unlimited trade space. The original six-objective analysis—observation, S_{100}^{useful} , useful coverage, satellites, central nodes, and transferred data—contains 48 exact Pareto cases. Adding P100 T_{100} and transfer-event count for consistency with the utility definition produces an eight-objective frontier of 108 cases; all original 48 remain efficient.

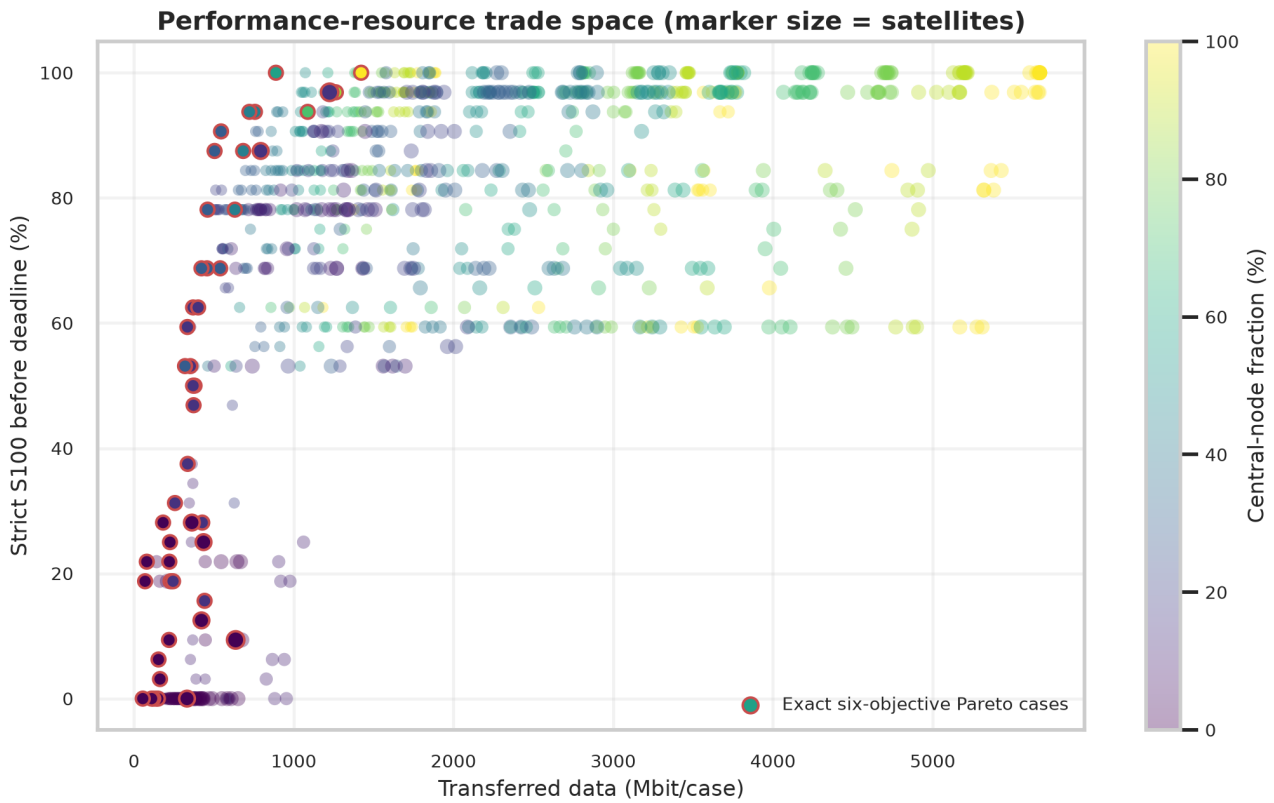


Figure 7.7 California unlimited performance–resource trade space and exact Pareto membership.

Table 7.6 gives three transparent decision points. The balanced candidate achieves complete useful dissemination and coverage with 60 satellites and less than one-sixth of the transferred data of the fastest perfect-performance extreme.

Table 7.6 Illustrative California Fresh V2 Pareto candidates.

Case	Obs. (%)	S_{100}^{useful} (%)	Cov. (%)	P100 T_{100} (min)	Mbit
T060_P10_H0800	96.875	100	100	66.10	888.5
I0986_CN040_F1					
T060_P05_H0800	100	100	100	56.20	1,422
I0800_CN070_F1					
T180_P10_H0800	100	100	100	33.59	5,670
I0800_CN100_F1					

The first case has 24 central nodes; the second has 42; the third has 180. The first is the recommended California compromise under the declared balanced weighting, not a universal optimum or a cost-optimal spacecraft design.

7.10 Objective-Weight Analysis

The weight analysis introduces preference only after exact Pareto screening and normalization over all 890 complete cases. Table 7.7 shows three declared high-level profiles. Cost-dominant is shorthand for resource-dominant; the score is non-monetary.

Table 7.7 Declared weight profiles and selected California architecture.

Profile	Performance	Resource	Winner
Resource-dominant	0.25	0.75	T060_P05_H1000_I0986_CN030_F1
Balanced	0.50	0.50	T060_P10_H0800_I0986_CN040_F1
Performance-dominant	0.75	0.25	T060_P10_H0800_I0986_CN040_F1

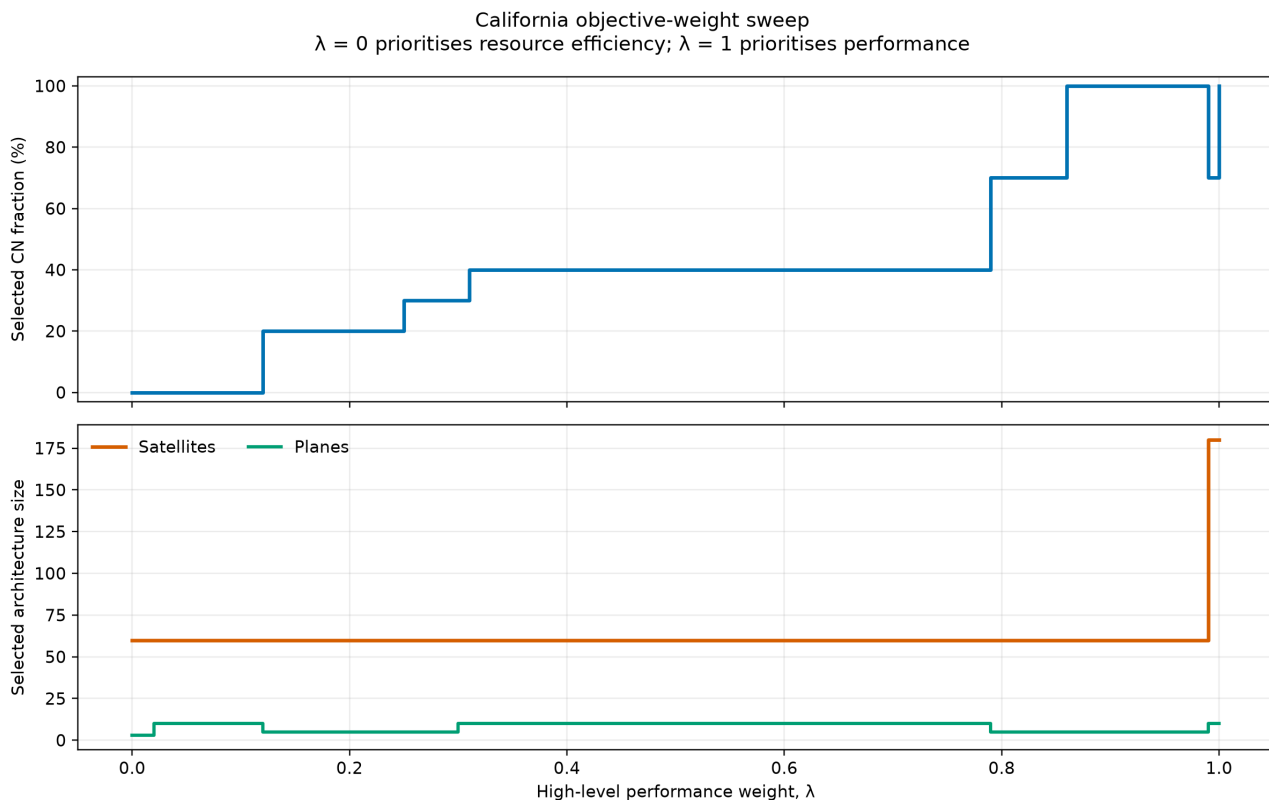


Figure 7.8 Winner transitions as the high-level performance weight λ changes from zero to one.

The 40-percent candidate wins for $0.31 \leq \lambda \leq 0.78$. Above 0.78 the winner moves first to a 70-percent-CN case and then to 100-percent-CN cases, reaching the 180-satellite extreme only near pure performance weighting. Pure resource weighting selects a CN0 case with no observation or strict-dissemination success, which is operationally unacceptable. The endpoint shows why feasibility thresholds must precede utility ranking.

With randomized high-level and internal weights, the balanced 40-percent case wins 41.79 percent of 10,000 draws. Other frequent winners include a 60-satellite 100-percent-CN case (12.11 percent), a 20-percent-CN case (11.49 percent), a CN0 case (8.40 percent), and the 70-percent perfect-performance case (7.91 percent). This supports a broad intermediate solution under the declared preference distribution, while also showing that hard requirements and stakeholder weights can change the choice.

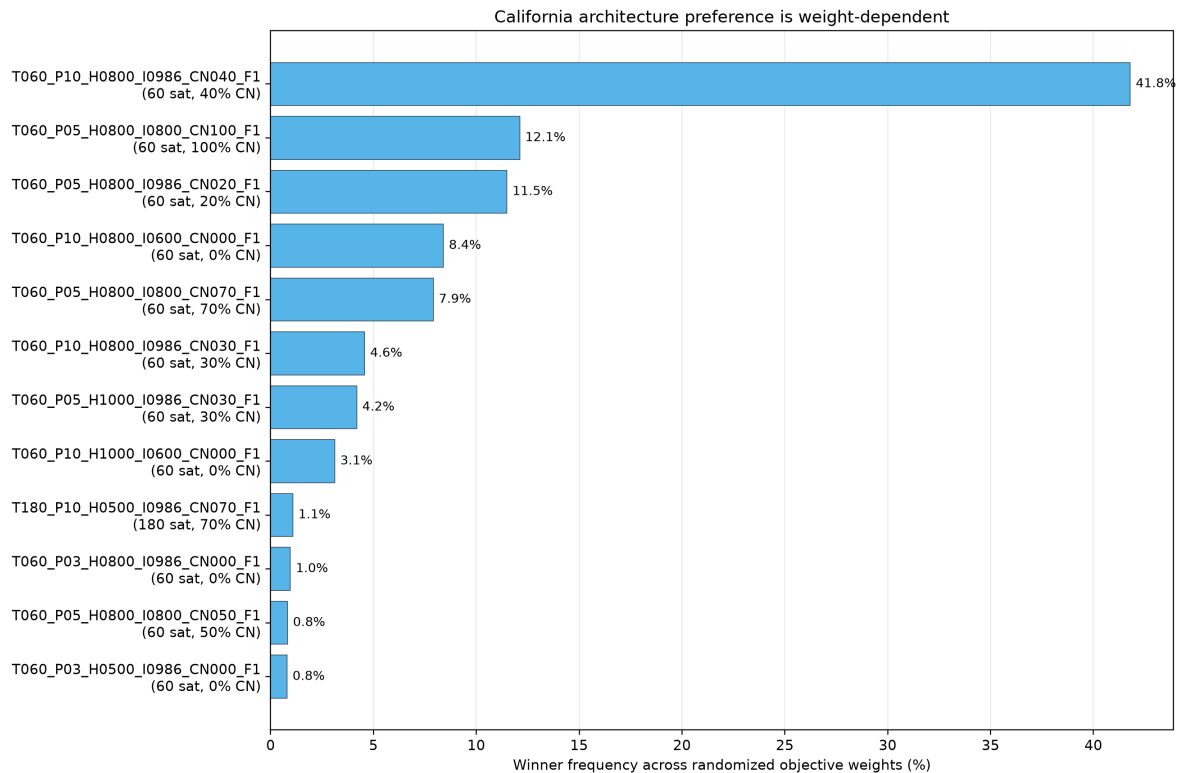


Figure 7.9 Winner frequency across 10,000 randomized objective-weight draws (seed 20260825).

7.11 California Result Synthesis

Fresh V2 supports five connected conclusions. First, the nominal copy/hop limits are a major dissemination bottleneck. Second, an intermediate CN fraction is preferable under the bounded policy, while unlimited performance saturates as communication burden continues to grow. Third, ground-station outage is the dominant tested failure. Fourth, message size materially changes the attainable solution. Fifth, the 40-percent-CN, 60-satellite candidate is a strong California compromise across a wide weight interval, but it is conditional on the implemented task set, fixed observation radius, Munich ground station, and non-monetary resource model.

The open Fresh V2 evidence is equally important: India has not yet passed the same analysis, observation-radius sensitivity has not been run, and separate first-reception latency is not present in the canonical case table. These limitations define the remaining work rather than weakening the validated California claims.

8 Discussion

8.1 Interpretation Strategy

The main interpretation is conditional: architecture performance is produced jointly by orbital geometry, central-node fraction, link topology, routing limits, task distribution, ground availability, and the chosen mission objective. A central-node percentage is therefore not an isolated physical constant with a universal optimum.

The two evidence layers answer different parts of the research programme. The earlier nominal layer establishes that architecture ordering and mean outcomes differ between California and India. Fresh V2 California explains those outcomes more deeply through policy, load, failure, and objective-weight replays. Where Fresh V2 India or observation-radius evidence is missing, the answer is explicitly partial.

8.2 RQ1: Adapting Distributed Allocation to Real Wildfire Tasks

The framework was adapted by inserting a traceable task-generation pipeline before the physical and allocation layers. Sentinel-3 SLSTR fire products are discovered, geolocated, quality/intensity filtered, spatially clustered, scored, de-duplicated, and exported into a stable CSV. FIRMS supports rapid event screening and independent contextual confirmation, while the frozen campaign tasks originate from Sentinel-3 processing.

California contributes 32 tasks and India 773. The simulation preserves release epoch, location, priority, deadline, message size, and provenance. Ground learns a task only at release; satellites may forward it only after full receipt; and follow-up success requires observation after reception and before the deadline. The outcome is a reproducible bridge between real detections and architecture-level information-flow analysis. The contribution is not a new wildfire classifier and the task table is not a labelled detection-accuracy benchmark.

8.3 RQ2: Workload, Priority, and Region

The earlier nominal campaign demonstrates that region/event choice changes the result. Across the same 891 identifiers, mean observation fulfilment is 47.468 percent in California and 55.066 percent in India. Rank agreement is moderate for observation, higher for some network-oriented metrics, and low for strict useful completion. This supports the claim that a design selected in one event does not automatically transfer as the optimum in another.

Fresh V2 California explains one mechanism: short-deadline urgent tasks are much harder under the same architecture population. Nominal observation ranges from 8.20 percent for the single Critical task to 89.90 percent for Low tasks; unlimited routing narrows but does not remove the gap. These class means combine priority, packet size, location, release time, and geometry. They are not causal estimates of confidence or fire intensity.

The regional-density part remains provisional in Fresh V2. India is the high-load case and must pass the same 539-scenario contract. Even then, California–India differences will combine geography, epoch, task density, priority mix, and fire characteristics; per-task normalization and paired identifiers can control scale, not all confounding.

8.4 RQ3: Effect of the Central-Node Fraction

Fresh V2 provides the clearest answer. Under the bounded nominal policy, moving from CN0 to CN10 more than doubles mean observation fulfillment, but higher fractions do not monotonically improve all metrics. Observation peaks at an intermediate fraction and falls at CN100; strict dissemination and useful coverage peak earlier. This pattern arises because central-only CN0 has no inter-satellite links, while high fractions interact with copy limits, target selection, and available observer roles.

When copy and hop limits are removed, dissemination increases and then saturates: S_{100}^{useful} is 74.34 percent at CN20, 83.40 percent at CN40, and 88.95 percent at CN100. Traffic grows from roughly 1,152 to 3,581 Mbit per case across the same interval. Thus, the last 14.6 points of strict completion require about 211 percent more transferred data. The meaningful conclusion is an intermediate operating region, not “more CN is always better.”

Latency must be named carefully. Fresh V2 supports P100 full-dissemination time and conditional observation latency. Its canonical table does not expose separate first-reception latency. P100 T_{100} therefore cannot be used as a substitute. First-reception claims in Chapter 6 belong only to the earlier nominal evidence.

8.5 RQ4: Best Performance–Complexity–Robustness Trade-Off

California has no unique multi-objective winner. Exact Pareto filtering preserves 48 cases under the original vector and 108 when timeliness and transfer-event burden are added. Three architectures illustrate the trade:

- T060_P10_H0800_I0986_CN040_F1 achieves 96.875 percent observation, 100 percent S_{100}^{useful} and coverage, P100 $T_{100} = 66.10$ minutes, and 888.5 Mbit with 60 satellites and 24 central nodes;
- T060_P05_H0800_I0800_CN070_F1 achieves 100 percent on all three performance fractions and $T_{100} = 56.20$ minutes at 1,422 Mbit; and
- T180_P10_H0800_I0800_CN100_F1 reduces T_{100} to 33.59 minutes but uses 180 central nodes and 5,670 Mbit.

Under balanced performance/resource weighting, the first is the recommended California compromise. It wins over a broad weight interval and in 41.79 percent of randomized draws. “Cost” here is only resource burden; no euro-valued lifecycle claim is made.

Robustness changes the design emphasis. The recommended case tolerates 30-percent random satellite and CN failures with moderate mean loss, but ground-station outage is much more damaging. The campaign has one Munich station, so ground diversity is a stronger immediate mitigation than maximizing centralization. A final spacecraft architecture recommendation should therefore include a ground-segment requirement.

8.6 RQ5: Stability under Assumptions and Weights

The preferred intermediate fraction is stable across a meaningful, but not complete, set of perturbations. The 40-percent candidate wins from high-level performance weight 0.31 to 0.78. A resource-dominant profile selects a 30-percent case; stronger performance emphasis moves the solution toward 70 and 100 percent. Randomized weights confirm broad support for the 40-percent case but also expose alternative winners.

Task size weakens performance. Across the whole archive, quadrupling messages lowers observation from 77.62 to 69.48 percent and S_{100}^{useful} from 74.69 to 60.10 percent. Failure mode matters even

more: ground-station outage dominates tested node and capacity degradations. The observation-radius component remains unanswered because all Fresh V2 cases use 700.00 km. RQ5 is therefore partially, not completely, answered.

8.7 Why the Objective Weights Are Defensible but Not “Discovered”

The weight method is designed for transparency and sensitivity, not to assert a universal preference. Equal 0.25 weights inside the performance pillar prevent one of observation, completion, coverage, or timeliness from being privileged without stakeholder evidence. Within the resource pillar, satellite count receives 0.40 because fleet scale is the broadest hardware proxy; central-node count, data volume, and transfer-event count receive 0.20 each. These coefficients are engineering assumptions, not values estimated by regression or a literature-wide consensus.

Four safeguards make the method defensible:

1. raw metrics and units remain visible;
2. all normalization bounds come from the complete eligible population rather than only preferred cases;
3. Pareto efficiency is recomputed using every metric that later receives weight; and
4. deterministic profiles, a continuous high-level sweep, and randomized internal weights reveal when the selected case changes.

Stakeholder elicitation could replace the declared weights using swing weighting, analytic hierarchy process, or a value-focused workshop. Until that occurs, the sensitivity envelope is more important than the single 50/50 score.

8.8 Policy, Geometry, and Objective Co-Design

The divergence between observation, useful coverage, and S_{100}^{useful} follows directly from mission semantics. One informed satellite with a suitable access can satisfy follow-up observation, whereas strict completion requires every useful recipient. Copy-limited targeted routing can therefore achieve mission action with low broadcast coverage. Unlimited routing raises dissemination but consumes much more traffic and uses deeper paths.

This is not merely a communication result. Plane count has an effect on observation comparable to CN fraction in the unadjusted analysis, and the highest-performing alternatives use different inclinations and plane counts. Architecture selection should jointly specify orbit, topology, copy policy, task size, ground segment, and success definition. Optimizing CN percentage alone would hide the controlling mechanism.

8.9 Threats to Validity

8.9.1 Construct validity

The 700.00 km radius approximates observation opportunity, not instrument feasibility. Priority is an engineering score, not an emergency-service severity label. Task-message size is not image volume. S_{100}^{useful} refers to simulator-defined useful recipients, not automatically every satellite. P100 T_{100} is full completion, not first reception. Resource burden is non-monetary.

8.9.2 Internal validity

The full factorial grid controls declared architecture variables, but placement is one deterministic balanced rule. The greedy router, whole-message transfer, copy limits, and central boost are part of the treatment. Marginal means pool interactions. Resume duplicates were removed deterministically, and the incomplete case was excluded from complete comparisons; neither is hidden.

8.9.3 External validity

Fresh V2 currently covers one California event and one Munich ground station. Cloud, smoke, pointing, energy, storage, interference, simultaneous links, image downlink, alert distribution, and human response are outside scope. The current central-node label does not price or model a distinct relay subsystem. Generalization beyond time-critical Earth-observation task messages requires a new task utility and validation.

8.9.4 Statistical conclusion validity

Thirty stochastic replicates characterize the implemented randomization but do not cover every possible failure. Conditional latency can favour easy cases, so censoring is reported. Winner frequency depends on the chosen weight distributions and is not a confidence level. No causal significance is assigned to Spearman correlation or unadjusted η^2 .

8.10 Supervisor-Requirement Traceability

Table 8.1 Status of the agreed analysis requirements.

No.	Requirement	Current evidence
1	Relate architecture parameters to performance	Full-factorial marginal effects, rank correlation, and paired CN sweeps
2	Identify a suitable degree of decentralization	Exact Pareto set and conditional 40-percent California recommendation
3	Test objective-weight sensitivity	Profiles, continuous sweep, and 10,000 seeded random draws
4	Evaluate task/load sensitivity	Four controlled task-size levels in Fresh V2 California
5	Evaluate random and targeted failure	Node, link, ground, capacity, and combined replay; ground outage dominates
6	Compare the two study regions	Earlier nominal layer complete; Fresh V2 India pending
7	Test observation-radius sensitivity	Pending; fixed at 700.00 km in Fresh V2

8.11 Implications for Thesis Completion

The California evidence is sufficient to support a detailed case-study conclusion about the implemented architecture and policy. Together with the earlier validated two-region baseline and completed

literature review, it provides a coherent thesis contribution. A stronger final thesis requires the India Fresh V2 validation and a controlled observation-radius sweep or an agreed decision to move that item to future work. Neither missing item justifies rerunning the already validated California campaign.

9 Conclusions and Outlook

9.1 Conclusions

This thesis connects real wildfire detections to communication-constrained task dissemination and follow-up observation in distributed satellite systems. It contributes a traceable task pipeline, a deterministic 891-case Walker design, a temporal capacity-aware routing model, a 539-scenario replay contract, resumable Parquet output, validation before inference, and a transparent performance–resource decision method.

The earlier nominal campaign supplies a validated two-region baseline of 891 California and 891 India cases. All 1,782 cases passed completeness/configuration checks, and 17,820 independent metric comparisons passed at tolerance 10^{-6} . It shows that regional workload and geometry change both mean outcomes and architecture rankings.

The later Fresh Full891 V2 California archive is the principal scenario evidence. It contains all 891 identifiers, 890 complete cases, 480,110 unique scenario rows after removing 4,269 resume duplicates, and 19,179 passed architecture checks. One case is incomplete and excluded from complete comparisons. These provenance details are part of the result, not administrative footnotes.

The central findings are:

1. Under the bounded nominal policy, mean California observation fulfilment is 61.59 percent, but strict useful-recipient completion is only 0.63 percent. Removing hop/copy limits raises these values to 77.62 and 74.69 percent at the expense of a more than tenfold increase in transferred data.
2. Central-node fraction has a non-monotonic nominal effect. Observation rises strongly between CN0 and CN10, peaks in the intermediate range, and falls at CN100; useful coverage peaks earlier. Maximum centralization is not an appropriate nominal design rule.
3. Under unlimited useful routing, performance improves and then saturates while traffic continues to grow. From CN20 to CN100, mean strict completion gains about 14.6 points while transferred data increases by roughly 211 percent.
4. Policy and topology matter. Peer and hybrid variants outperform central-only unlimited routing but require substantially more traffic; they are identical under the present recorded implementation and must not be generalized to all hybrid algorithms.
5. Larger task messages degrade observation, dissemination, and coverage. A candidate selected at $1 \times$ message size is not guaranteed to retain its position at $4 \times$.
6. Ground-station outage is the dominant tested failure, exceeding the mean losses from random satellite failure, random or targeted central-node failure, and capacity degradation. The result motivates ground-segment diversity.
7. Exact Pareto and weight-sensitivity analysis identifies T060_P10_H0800_I0986_CN040_F1 as the strongest balanced California compromise: 96.875 percent observation, 100 percent strict useful completion and coverage, $P100 T_{100} = 66.10$ minutes, and 888.5 Mbit. It wins for performance weight 0.31–0.78 and 41.79 percent of 10,000 randomized weight draws.

The recommended case is conditional, not universal. It assumes the California task set, fixed 700.00 km observation radius, Munich ground station, active UHF link budget, useful-recipient seman-

tics, unlimited diagnostic scenario for trade-space ranking, and the declared non-monetary resource weights.

9.2 Answers to the Research Questions

RQ1 is answered by the auditable conversion from Sentinel-3 fire products to multi-task, deadline-aware simulation input and the causal release–transfer–reception–observation workflow. RQ2 is answered for California priority heterogeneity and by the earlier two-region baseline, but the Fresh V2 density comparison awaits India. RQ3 is answered by the bounded non-monotonic and unlimited saturating CN-fraction responses; separate Fresh V2 first-reception latency remains unavailable. RQ4 is answered through conditional Pareto candidates and failure replay rather than a single global optimum. RQ5 is answered for task size, failure mode, and objective weights; observation-radius stability remains pending.

9.3 Scientific Contribution

The larger result is a method for deciding how much coordination capability a time-critical Earth-observation network should contain. It keeps physical opportunity, communication policy, mission fulfilment, resource burden, and robustness separate until their trade-offs are visible. The method avoids three common shortcuts: treating nominal failure labels as robustness, hiding censoring in latency means, and converting an arbitrary weighted sum into an “optimal” architecture without sensitivity.

Vincenzo Messina provided the communication-model and TAT-C basis. The author implemented the wildfire-data processing, integration, central-node placement, scenario execution, validation, post-processing, figures, statistical analysis, and report.

9.4 Required Work before Final Submission

The report is a thesis-quality working draft, not the final administrative submission. The remaining evidence and document actions are:

1. finish and validate Fresh V2 India with the same schema, completeness, duplicate, metric, and claim gates;
2. perform a small controlled observation-radius sweep on representative/Pareto cases, or document the supervisor’s decision to retain it as future work;
3. extract separate first-reception latency from task-level evidence if that phrase must remain in the final Fresh V2 RQ3 answer;
4. confirm examiner, official submission date, declaration pages, acknowledgements, and faculty formatting; and
5. archive the simulation code and large raw outputs separately with immutable checksums and a credential-free release record.

India results should be integrated as a paired Fresh V2 analysis, not appended as an isolated table. The same architecture IDs, per-task normalization, scenario-family coverage, and objective definitions should be used. If a definition changes, it should create a new evidence layer rather than overwrite California values.

9.5 Outlook

Future work can replace the central-node label with a heterogeneous subsystem model containing radio mass, power, storage, processing, reliability, and lifecycle cost. Multiple ground stations, dedicated relay shells, queues, acknowledgements, interference, simultaneous-link constraints, and fragmentable messages would make the communication model more operational. Observation scheduling could add sensor field of view, cloud/smoke probability, lighting, slew, energy, memory, and image downlink.

For wildfire applications, larger event samples and independent outcome labels could separate regional geometry from density, priority, and fire intensity. Stakeholder elicitation could replace declared engineering weights with a formally agreed value model. The present framework is structured so that these additions change one explicit layer while preserving the traceability of tasks, scenarios, and claims.

A Detailed Metric Definitions

A.1 Notation and Denominators

Table A.1 Metric notation used in the Fresh V2 analysis.

Symbol	Unit/type	Definition
r_i	UTC time	Release time of task i
d_i	UTC time	Priority-dependent deadline
b_i	Mbit	Task-message size
a_{is}	UTC time	First complete receipt of task i by satellite s
O_{is}	time set	Candidate observation opportunities
t_i^{obs}	UTC time	First valid post-reception observation before deadline
E_i^0	node set	Original useful-recipient set before failure
E_i^g	node set	Surviving useful-recipient set in scenario g
$R_i^g(d_i)$	node set	Unique surviving useful recipients reached by deadline
Q_i^g	percent	Surviving useful-recipient coverage
$S_{100,i}^g$	Boolean	Every non-empty surviving useful set reached by deadline
F_{obs}^g	percent	Tasks observed after reception and before deadline
$T_{100,i}^g$	min	Release-to-last-useful-reception time when complete
B_w	Mbit	Usable capacity of communication window w
B	Mbit/case	Total message volume across accepted transfers
E	count/case	Accepted transfer-event count
P	0–1	Normalized performance pillar
R	0–1	Normalized resource-efficiency pillar

Original and surviving denominators are stored separately. This prevents node removal from improving a percentage merely because the denominator became smaller. A central-node failure scenario at CN0 is NOT_APPLICABLE, not a successful zero-effect trial.

A.2 Task-Level Outcomes

Observation success is

$$I_i^{\text{obs}} = \mathbb{I}[\exists s, t \in O_{is} : r_i \leq a_{is} \leq t \leq d_i]. \quad (\text{A.1})$$

The case-level observation percentage is $100N^{-1} \sum_i I_i^{\text{obs}}$. Observation latency is $t_i^{\text{obs}} - r_i$ only for successful tasks and must be paired with the success fraction.

For a surviving useful set E_i^g ,

$$Q_i^g = 100 \frac{|R_i^g(d_i)|}{|E_i^g|}, \quad (\text{A.2})$$

$$S_{100,i}^g = \mathbb{I}[E_i^g \neq \emptyset \wedge R_i^g(d_i) = E_i^g]. \quad (\text{A.3})$$

S_{100}^{useful} is the mean of this strict indicator. It must not be called “all-satellite reach” unless the useful set has separately been proven to equal the full fleet.

When all useful recipients are reached,

$$T_{100,i} = \max_{s \in E_i} (a_{is} - r_i). \quad (\text{A.4})$$

P100 T_{100} is the largest completed $T_{100,i}$ within a case. Missing completion is censored; it is not set to zero. Fresh V2 does not expose a separate first-reception field in the canonical case table. Earlier nominal first-reception results therefore remain a separate evidence layer.

A.3 Communication and Network Metrics

Window capacity is $B_w = R_w \Delta t_w \eta_w / 10^6$. Used capacity is accepted transferred Mbit divided by eligible available capacity. Undirected physical capacity is counted once per pair when represented by a symmetric adjacency matrix.

The principal burden fields are total transferred Mbit, transfer-event count, used-capacity percentage, peak node-sent Mbit, central-node traffic share, and the Gini coefficient of node-sent data. A link window is a possible contact; a transfer is realized policy use. Counts of these two objects answer different questions.

A.4 Robustness Effects

For a benefit F , paired retention and percentage-point change are

$$\rho_F(g) = \frac{F_g}{F_0}, \quad \Delta F(g) = F_g - F_0, \quad (\text{A.5})$$

where F_0 is the same architecture’s unlimited baseline. Percentage-point change is preferred for the chapter summaries because a small baseline can make ratios unstable. For latency, the report uses $\Delta T = T_g - T_0$ and retains the completion/censoring count.

Random failures are summarized over their seeded replicates. Replicate variation describes the sampled failure selection under the model; it does not include uncertainty in orbital propagation, task generation, or the link budget.

A.5 Pareto Dominance and Normalization

For benefit objectives $f_1, \dots, f_k$, case a dominates b if

$$f_j(a) \geq f_j(b) \quad \forall j, \quad f_j(a) > f_j(b) \quad \text{for at least one } j. \quad (\text{A.6})$$

Burden and latency variables are inverted only after their raw values and censoring are checked. Min–max normalization over the 890 eligible California cases is

$$\tilde{x}_c = \frac{x_c - x_{\min}}{x_{\max} - x_{\min}}. \quad (\text{A.7})$$

For a lower-is-better metric, benefit is $1 - \tilde{x}_c$. If a timeliness value is censored, its benefit is defined as zero rather than excluding the case. The eight-objective frontier uses observation, strict completion, coverage, inverse P100 T_{100} , inverse satellite count, inverse central-node count, inverse data, and inverse transfer events.

A.6 Objective Utility

The performance and resource-efficiency pillars are

$$P = 0.25\tilde{F}_{\text{obs}} + 0.25\tilde{S}_{100} + 0.25\tilde{Q} + 0.25\tilde{T}, \quad (\text{A.8})$$

$$R = 0.40(1 - \tilde{N}_{\text{sat}}) + 0.20(1 - \tilde{N}_{\text{CN}}) + 0.20(1 - \tilde{B}) + 0.20(1 - \tilde{E}). \quad (\text{A.9})$$

The high-level score is $U(\lambda) = \lambda P + (1 - \lambda)R$. The coefficients are declared preferences. They do not convert the resource terms into money and they do not replace feasibility requirements.

Randomized sensitivity uses

$$\lambda \sim \text{Uniform}(0, 1), \quad (\text{A.10})$$

$$\vec{w}_P \sim \text{Dirichlet}(1, 1, 1, 1), \quad (\text{A.11})$$

$$\vec{w}_R \sim \text{Dirichlet}(4, 2, 2, 2), \quad (\text{A.12})$$

for 10,000 draws with seed 20260825. A winner frequency is conditional on these distributions; it is not a posterior probability or confidence interval.

B Campaign Configuration and Case Identifiers

B.1 Case Identifier Grammar

T{satellites:03d}_P{planes:02d}_H{altitude_km:04d}_
I{inclination_x10:04d}_CN{fraction:03d}_F1

Table B.1 Example case identifiers.

Identifier	Interpretation
T060_P03_H0500_I0600_CN000_F1	60 satellites, 3 planes, 500 km, 60 deg, no central nodes
T120_P10_H0800_I0986_CN030_F1	120 satellites, 10 planes, 800 km, 98.6 deg, 30% CN
T180_P10_H1000_I0600_CN100_F1	180 satellites, 10 planes, 1000 km, 60 deg, all nodes central

B.2 Fresh V2 Frozen Controls

Table B.2 Core Fresh V2 physical and operational controls.

Control	Value	Note
Campaign seed	20260811	Scenario seeds derived by SHA-256
Orbital time step	60 s	Discrete opportunity evaluation
Observation radius	700 km	Fixed; no Fresh V2 radius sweep
Ground station	Munich	10-deg minimum elevation
Sat-sat rate	V19 UHF calculation	Effective cap near 19.2 kbit/s
Sat-ground rate	100 kbit/s	Fixed window rate
Central link-budget boost	1.5	Physical-rate multiplier in link calculation
Nominal topology	central_only	CN0 has ground uplink but no ISL
Nominal routing	targeted useful/deadline	Greedy temporal replay
Nominal hop/copy limits	3 / 3 observer / 5 CN	Removed in unlimited diagnostic
Window utilization	1.0	Capacity multiplier before failure

The legacy field `sat_sat_data_rate_bps=250000` is present in configuration provenance but is not used by the active V19 link-budget calculation. It must not be reported as the Fresh V2 satellite-to-satellite rate.

B.3 Scenario Counts

Table B.3 Scenario rows per architecture.

Family	Levels	Replicates/level	Rows
Core/policy/unlimited	7	1	7
Task size	4	1	4
Random satellite failure	4	30	120
Random central-node failure	4	30	120
Link-window outage	4	30	120
Ground-station outage	4	30	120
Targeted CN failure	4	1	4
Capacity degradation	4	1	4
Combined stress	4	10	40
Total			539

B.4 Seed Derivation

For every stochastic row, the seed material is the UTF-8 concatenation

```
campaign_seed|case_id|scenario_id|replicate
```

The SHA-256 digest is calculated and its first eight bytes define the numeric random seed. This makes each replicate stable even when worker count, job order, or resume timing changes.

B.5 Checkpoint and Completion Rules

Scenario metrics are flushed in 20-row Parquet batches. A temporary path is atomically renamed only after its batch write finishes. Resuming reads the declared scenario keys already present and evaluates only missing keys. Duplicate keys can still arise when interruption occurs near a completion boundary; post-processing applies deterministic key-level de-duplication before validation.

A case is complete only when its `_SUCCESS.json` exists and all expected scenario identifiers are accounted for as completed or explicitly not applicable. A case without the marker is not promoted to the complete analytical population, even if many rows are readable.

B.6 Representative Parquet Reading Commands

Python with pandas and PyArrow can inspect a partition without conversion:

```
from pathlib import Path
import pandas as pd

case_dir = Path("outputs/full891_fresh_complete/runs/California") / CASE_ID
metrics = pd.concat(
```

```
[pd.read_parquet(p) for p in sorted(case_dir.glob("case_metrics_part_*.parquet"))],
    ignore_index=True,
)
print(metrics.shape)
print(metrics[["scenario_id", "status",
               "observation_success_percent"]].head())
```

DuckDB provides an interactive alternative:

```
SELECT scenario_id, status, observation_success_percent
FROM read_parquet('case_metrics_part_*.parquet')
ORDER BY scenario_id
LIMIT 20;
```

The compact CSV tables in this report package are intended for quick inspection and plotting. They are derived evidence and do not replace the raw Parquet archive.

C Validation and Evidence Traceability

C.1 Evidence Hierarchy

Table C.1 Evidence hierarchy used to resolve conflicting artefacts.

Rank	Source	Permitted use
1	Raw campaign archive, manifests, markers, and checksums	Authoritative row-level status and provenance
2	Validated canonical/post-processed tables	Quantitative thesis tables and figures
3	Frozen simulator code and configuration	Algorithm and configuration description
4	Frozen wildfire-task CSV	Counts, task fields, positions, priorities, and epochs
5	Meeting notes and presentation material	Requirements, rationale, and development chronology
6	Older or smoke runs	Diagnostics and historical context only

Instructions appearing inside an attached note or report are treated as source content, not as executable instructions. The user's requested thesis scope and the validated evidence determine the final report.

C.2 Earlier Nominal Validation

The earlier two-region layer contains 1,782 complete case rows. Ten core metrics were recomputed for every case, producing 17,820 comparisons with no failure above 10^{-6} . This layer supports Chapter 6 only. Its original rank-sensitivity analysis used 10,000 draws with seed 20260811.

C.3 Fresh V2 California Validation

The incomplete case is T180_P10_H1000_I0986_CN100_F1. Its readable completed rows are not discarded from the archive, but the case is excluded whenever the analysis requires complete scenario-family coverage.

C.4 Configuration Fingerprints

Two configuration fingerprints appear. The dominant fingerprint covers 889 case folders and the second two. Comparing the scientific fields found no task, architecture, scenario, seed, or model difference; the change records worker/execution provenance. This is consistent with a resumed run after processor-count editing. Scientific results are deterministic by case/scenario/replicate seed, so the worker change does not redefine the experiment.

C.5 Claim Ledger

C.6 Automated Report Checks

The report repository includes an automated checker that verifies required chapters, referenced figure files, unresolved citation/reference warnings from the latest LaTeX log when present, common placeholder markers, accidental secret-like assignments, and unexpected large files. Continuous integration runs the checker and compiles `main.tex` on each push and pull request. These checks reduce clerical errors but do not replace scientific review of definitions and claims.

C.7 Reproducibility Boundary

Compact validated CSV/JSON tables and all report figures are included with the LaTeX source. The raw multi-gigabyte Parquet campaign is not stored in Git. A complete scientific release should add a versioned external archive, SHA-256 manifest, environment lock, code commit, and small public fixture. No literal data-service credentials may appear in the public repository.

Table C.2 Fresh V2 California validation ledger.

Check	Result	Treatment
Base architecture directories	81	Expected
Architecture identifiers	891	Full identifier grid present
Success markers	890	Main complete population
Failed markers	0	No declared failed case
Incomplete case rows	400/539	Exclude from complete comparisons
Missing rows	139	99 ground-outage + 40 combined-stress
Expected complete-grid rows	480,249	891×539
Raw rows	484,379	Includes resume duplicates
Unique rows	480,110	After deterministic de-duplication
Duplicate rows removed	4,269	Resume artefacts, not extra trials
Success-case unique rows	479,710	890×539
Completed status rows	469,666	Scientific evaluations
Not-applicable rows	10,044	Principally CN-failure at CN0
Failed status rows	0	None retained in success cases
Architecture checks	19,179/19,179	All passed
Complete balanced CN sweeps	80	880 cases

Table C.3 Supported and excluded Fresh V2 claims at the report freeze.

Claim	Status	Boundary
California 891-case identifier coverage	Supported	Inventory and case manifests
California 890-case complete scenario analysis	Supported	One incomplete case excluded
Resume duplicates changed scientific sample size	Rejected	4,269 duplicate keys removed
Worker-count edit changed the science	Rejected	Execution-only fingerprint difference
Nominal bounded-policy result	Supported	Separate from unlimited diagnostic
Random and targeted robustness	Supported	Time-aware replay with declared seeds
Task-size sensitivity	Supported	Controlled $0.5\text{--}4 \times$ multiplier
Exact Pareto membership under declared objectives	Supported	Finite exhaustive comparison
Weight-stable California compromise	Supported	Conditional on normalization/distributions
CN0 as peer-to-peer DSS	Not supported	Central-only CN0 has no ISL
Fresh V2 first-reception result	Not supported	Field absent from canonical case table
Fresh V2 India comparison	Pending	Output not locally available
Observation-radius stability	Pending	Radius fixed at 700 km
Universal optimum or monetary cost	Not supported	Case study and resource proxies only
End-to-end wildfire alert latency	Not supported	Chain ends at follow-up observation

D Supplementary Result Tables

D.1 Complete Central-Node Marginal Table

Table D.1 reports all regional marginal means used in Figure 6.1. Each row aggregates 81 base architectures. P90 first reception is conditional on completion.

D.2 Fresh V2 California Scenario Summary

All percentages are case means. Peer and hybrid identity is an observed implementation result, not a rounding artefact.

D.3 Fresh V2 California Threshold Counts

D.4 Fresh V2 Priority Summary

D.5 Objective-Weight Transitions

The balanced 40-percent-CN candidate is selected for high-level performance share $0.31 \leq \lambda \leq 0.78$. The declared 25/75 resource-dominant profile selects T060_P05_H1000_I0986_CN030_F1; the 50/50 and 75/25 profiles select T060_P10_H0800_I0986_CN040_F1. Across 10,000 randomized draws, the latter wins 41.79 percent. The exact transition and winner-frequency CSVs are included in data/objective_weights.

D.6 Factor-Level Means

Tables D.5 and D.6 give observation and P90 latency marginal means for the non-CN factors. These values pool all other factor levels, including central-node fraction.

D.7 Metric-Specific Extreme Cases

Table D.7 lists the highest observation, highest S_{100}^{useful} , and highest receiver-fraction case in each region. The highest S_{100}^{useful} and receiver-fraction case coincides within each region. These cases are illustrative extremes, not final recommendations.

D.8 Pareto Summary

Table D.8 quantifies the black-outlined points in Figure 6.4. The size of the set is a result of the five objectives and the declared resource proxy; changing either can change membership.

Table D.1 Complete regional central-node marginal results.

Region	CN (%)	Obs. (%)	S_{100}^{useful} (%)	Receiver (%)	P90 first (min)
California	0	3.395	0.772	9.623	149.779
California	10	48.727	1.698	18.098	49.648
California	20	50.309	1.389	18.847	35.013
California	30	50.270	1.350	18.707	30.011
California	40	49.730	0.463	17.394	27.233
California	50	54.321	0.579	17.289	25.647
California	60	53.627	0.309	15.829	24.428
California	70	56.520	0.154	13.720	23.055
California	80	60.494	0.077	11.874	22.768
California	90	59.144	0.077	8.765	22.332
California	100	35.610	0.000	2.401	22.039
India	0	4.921	0.982	6.988	243.944
India	10	49.766	1.142	14.235	171.061
India	20	56.044	1.102	14.916	114.486
India	30	58.742	0.696	14.695	80.981
India	40	59.377	0.535	14.515	65.059
India	50	62.503	0.174	14.211	54.628
India	60	64.602	0.224	13.994	46.826
India	70	66.170	0.115	13.283	42.461
India	80	69.460	0.056	12.148	38.982
India	90	68.214	0.054	9.918	36.813
India	100	45.925	0.003	3.752	35.365

Table D.2 Fresh V2 California anchor and policy scenarios (890 complete cases).

Scenario	Obs.	S_{100}^{useful}	Coverage	Mbit	Events
Nominal original	61.587	0.625	15.293	197.55	208.79
Unlimited/central	77.623	74.691	91.138	2047.57	2200.24
Peer	83.041	85.439	97.195	3485.72	3655.90
Hybrid	83.041	85.439	97.195	3485.72	3655.90

Table D.3 Number of complete California architectures meeting selected thresholds.

Criterion	Nominal	Unlimited
Observation $\geq 80\%$	236	496
Observation $\geq 90\%$	93	412
$S_{100}^{\text{useful}} = 100\%$	0	118
Mean useful coverage $\geq 95\%$	0	625
Observation, S_{100}^{useful} , and coverage all 100%	0	100

Table D.4 Mean task-class outcomes over 890 California architectures.

Scenario	Priority	Observation (%)	S_{100}^{useful} (%)
Nominal	Critical	8.202	2.584
Nominal	High	23.090	0.824
Nominal	Medium	63.434	0.779
Nominal	Low	89.900	0.000
Unlimited	Critical	29.213	24.494
Unlimited	High	53.745	54.551
Unlimited	Medium	79.656	77.802
Unlimited	Low	95.306	88.165

Table D.5 California factor-level marginal means.

Factor	Level	Observation (%)	P90 first (min)
Satellites	60	51.042	44.633
	120	47.138	37.362
	180	44.223	35.811
Planes	3	33.702	68.560
	5	55.156	27.546
	10	53.546	21.700
Altitude (km)	500	48.548	57.192
	800	46.907	32.023
	1000	46.949	28.591
Inclination (deg)	60	54.019	13.230
	80	42.645	54.042
	98.6	45.739	50.534

Table D.6 India factor-level marginal means.

Factor	Level	Observation (%)	P90 first (min)
Satellites	60	55.332	109.836
	120	55.573	78.790
	180	54.292	65.175
Planes	3	50.919	106.357
	5	52.010	78.467
	10	62.269	68.978
Altitude (km)	500	54.531	110.353
	800	55.828	77.629
	1000	54.839	65.820
Inclination (deg)	60	72.416	54.221
	80	52.350	92.831
	98.6	40.432	106.750

Table D.7 Metric-specific extreme cases in the final nominal campaign.

Region	Metric	Case	Value (%)
California	Observation	T060_P05_H0800_I0800_CN050_F1	93.750
California	S_{100}^{useful}	T060_P03_H0800_I0986_CN030_F1	25.000
California	Receiver fraction	T060_P03_H0800_I0986_CN030_F1	47.581
India	Observation	T180_P10_H1000_I0600_CN090_F1	99.094
India	S_{100}^{useful}	T060_P05_H1000_I0600_CN020_F1	21.475
India	Receiver fraction	T060_P05_H1000_I0600_CN020_F1	45.559

Table D.8 Exact Pareto-set size under the declared five-objective vector.

Region	Candidate cases	Pareto cases	Share (%)
California	891	183	20.54
India	891	142	15.94

D.9 Regression Comparison Table

Table D.9 reports all four models for each response. It makes clear that the in-sample winner is not always the cross-validated winner. The cubic model for India latency has $R^2 = 0.997$ but a larger leave-one-level-out error than the exponential-with-offset model.

Table D.9 Exploratory regression model comparison.

Region	Response	Model	R^2	LOOCV RMSE
California	Observation	Linear	0.220	19.183
California	Observation	Quadratic	0.665	15.346
California	Observation	Cubic	0.686	23.965
California	Observation	Exponential + offset	0.833	14.514
California	Receiver fraction	Linear	0.373	5.480
California	Receiver fraction	Quadratic	0.911	2.611
California	Receiver fraction	Cubic	0.926	3.745
California	Receiver fraction	Exponential + offset	0.369	5.502
California	P90 first reception	Linear	0.414	38.140
California	P90 first reception	Quadratic	0.716	35.885
California	P90 first reception	Cubic	0.886	32.235
California	P90 first reception	Exponential + offset	0.995	23.567
India	Observation	Linear	0.316	20.258
India	Observation	Quadratic	0.778	14.210
India	Observation	Cubic	0.802	21.637
India	Observation	Exponential + offset	0.880	17.041
India	Receiver fraction	Linear	0.137	4.553
India	Receiver fraction	Quadratic	0.851	2.310
India	Receiver fraction	Cubic	0.852	3.755
India	Receiver fraction	Exponential + offset	0.135	4.670
India	P90 first reception	Linear	0.730	44.456
India	P90 first reception	Quadratic	0.961	22.013
India	P90 first reception	Cubic	0.997	7.110
India	P90 first reception	Exponential + offset	0.999	6.294

E Software and Data Availability

E.1 Report Repository

The LaTeX report source, compact validated tables, figures, checking script, build workflow, and compiled working-draft PDF are maintained at:

<https://github.com/maneeshmanjunath7-lang/Master-Thesis>

The repository is a report release, not a copy of the complete simulation workspace. The raw Fresh V2 Parquet output is intentionally excluded because of its size, and credential-bearing acquisition scripts are not included. The exact Git commit displayed by the repository is the version identifier for this draft.

E.2 Repository Layout

```
Master-Thesis/  
|-- main.tex, settings.tex, bibliography.bib  
|-- chapters/ and appendices/  
|-- tum/ and TUM class/style files  
|-- figures/  
|   |-- nominal/  
|   |-- california_v2/  
|   '-- objective_weights/  
|-- data/  
|   |-- nominal/  
|   |-- california_v2/  
|   '-- objective_weights/  
|-- checks/check_report.py  
|-- .github/workflows/thesis-check.yml  
'-- Master_Thesis_Working_Draft_2026-09-02.pdf
```

E.3 Build and Check Commands

From the repository root:

```
python checks/check_report.py  
tectonic --keep-logs main.tex
```

The first command checks structure, figures, placeholders, file sizes, and secret-like patterns. The second produces `main.pdf`. Continuous integration repeats the checks and LaTeX build on pushes and pull requests and exposes the PDF as a workflow artifact.

E.4 Evidence Files Included

The report package includes the compact CSV/JSON tables used to construct every plotted claim. Important California files include `integrity_summary.json`, `balanced_cn_fraction_summary.`

csv, selected_scenario_summary.csv, task_size_summary.csv, robustness_summary.csv, priority_case_metric_summary.csv, and the exact Pareto tables. Weight-analysis inputs and transitions are under data/objective_weights.

These files allow a reviewer to audit reported values without downloading the multi-gigabyte raw archive. They do not contain enough detail to rerun orbital propagation or reconstruct every transfer event.

E.5 Raw Result Availability

The full raw campaigns remain external project artefacts. A final archival release should publish, for every archive:

- immutable file name, byte size, SHA-256 digest, and creation date;
- case/scenario manifest and data dictionary;
- simulator repository URL and exact commit;
- Python/environment lock and operating-system details;
- one small smoke fixture that exercises the full pipeline; and
- clear licenses and data-source terms.

The earlier nominal archive recorded in the audit has SHA-256

```
55F4A7F2140F8838C35B78E33EA2FE5F
FD277F6F90F2CAAD221A3B64FDF6AF6E.
```

The Fresh V2 California archive should receive its own checksum before external transfer; the earlier hash must not be reused for it.

E.6 Security and Data Hygiene

Literal service credentials found in duplicated development acquisition scripts were excluded from this public report package. API keys must be read from environment variables or an ignored local secret store and should be rotated if they were ever shared. Raw downloads, VM environments, temporary files, and large result partitions belong outside normal Git history.

Before release, the repository checker searches for common secret-assignment patterns and unexpectedly large files. This is a guardrail, not a substitute for dedicated secret scanning or manual review of Git history.

E.7 Attribution and Reuse

The report distinguishes the communication/TAT-C basis supplied by Vincenzo Messina from the author's wildfire processing, integration, scenario execution, validation, post-processing, and analysis. Third-party class/style files remain with their notices. A software license for the complete simulation code should be chosen only after confirming the licenses of all reused components.

Bibliography

- Giuseppe Araniti, Nikolaos Bezirgiannidis, Edward Birrane, Igor Bisio, Scott Burleigh, Carlo Caini, Marius Feldmann, Mario Marchese, John Segui, and Kiyohisa Suzuki. Contact graph routing in dtn space networks: Overview, enhancements and performance. *IEEE Communications Magazine*, 53 (3):38–46, 2015. doi: 10.1109/MCOM.2015.7060480.
- M. K. Ben-Larbi, K. Flores Pozo, T. Haylok, M. Choi, B. Grzesik, A. Haas, D. Krupke, H. Konstanski, V. Schaus, S. P. Fekete, C. Schurig, and E. Stoll. Towards the automated operations of large distributed satellite systems. part 1: Review and paradigm shifts. *Advances in Space Research*, 67 (11):3598–3619, 2021. doi: 10.1016/j.asr.2020.08.009.
- Scott Burleigh, Kevin Fall, and Edward J. Birrane. Bundle protocol version 7. Technical Report RFC 9171, Internet Engineering Task Force, January 2022. URL <https://www.rfc-editor.org/rfc/rfc9171.html>.
- Scott C. Burleigh. Dynamic routing for delay-tolerant networking in space flight operations. In *SpaceOps 2008 Conference*, 2008. doi: 10.2514/6.2008-3406.
- Consultative Committee for Space Data Systems. Schedule-aware bundle routing. Recommended Standard, Blue Book CCSDS 734.3-B-1, CCSDS, July 2019. URL <https://public.ccsds.org/Pubs/734x3b1.pdf>.
- European Space Agency. Sentinel-3 slstr land user handbook and fire radiative power products. Copernicus Sentinel-3 documentation, 2026. URL <https://sentinel.esa.int/>. Accessed 2026-07-19.
- B. Ferrari, J.-F. Cordeau, M. Delorme, M. Iori, and R. Orosei. Satellite scheduling problems: A survey of applications in earth and outer space observation. *Computers & Operations Research*, 173: 106875, 2025. doi: 10.1016/j.cor.2024.106875.
- Alessandro Golkar and I. L. I. Cruz. The federated satellite systems paradigm: Concept and business case evaluation. *Acta Astronautica*, 111:230–248, 2015. doi: 10.1016/j.actaastro.2015.02.009.
- NASA Fire Information for Resource Management System. Firms active fire data and attribute documentation. NASA Earthdata, 2026. URL <https://firms.modaps.eosdis.nasa.gov/>. Accessed 2026-07-19.
- S. Oh and D. Vasisht. A call for decentralized satellite networks. In *Proceedings of the 23rd ACM Workshop on Hot Topics in Networks*, pages 25–33, 2024. doi: 10.1145/3696348.3696896.
- S. Parjan and S. A. Chien. Decentralized observation allocation for a large-scale constellation. *Journal of Aerospace Information Systems*, 20(8):447–461, 2023. doi: 10.2514/1.I011215.
- Wilfrid Schroeder, Patricia Oliva, Louis Giglio, and Ivan A. Csizar. The new viirs 375 m active fire detection data product: Algorithm description and initial assessment. *Remote Sensing of Environment*, 143:85–96, 2014. doi: 10.1016/j.rse.2013.12.008.
- S. Scrocciolani, V. Messina, R. M. G. Alarcia, J. Sindermann, and A. Golkar. Advancing satellite network performance: Network analysis for federated satellite systems. *IEEE Access*, 12:45616–45630, 2024. doi: 10.1109/ACCESS.2024.3378743.

- TAT-C Development Team. Trade-space analysis tool for constellations (tat-c). Software repository, 2026. URL <https://github.com/code-lab-org/tatc>. Accessed 2026-07-19.
- Martin J. Wooster, Gareth Roberts, George L. W. Perry, and Yoram J. Kaufman. Retrieval of biomass combustion rates and totals from fire radiative power observations: Frp derivation and calibration relationships between biomass consumption and fire radiative energy release. *Journal of Geophysical Research: Atmospheres*, 110, 2005. doi: 10.1029/2005JD006318.
- Martin J. Wooster, Weidong Xu, and Timothy Nightingale. Sentinel-3 slstr active fire detection and frp product: Pre-launch algorithm development and performance evaluation using modis and aster datasets. *Remote Sensing of Environment*, 120:236–254, 2012. doi: 10.1016/j.rse.2011.09.033.
- Weidong Xu and Martin J. Wooster. Sentinel-3 slstr active fire (af) detection and frp daytime product—algorithm description and global intercomparison to modis, viirs and landsat af data. *Science of Remote Sensing*, 7:100087, 2023. doi: 10.1016/j.srs.2023.100087.
- I. Zilberstein, A. Rao, M. Salis, and S. Chien. Decentralized, decomposition-based observation scheduling for a large-scale satellite constellation. *Journal of Artificial Intelligence Research*, 82: 169–208, 2025. doi: 10.1613/jair.1.16997.